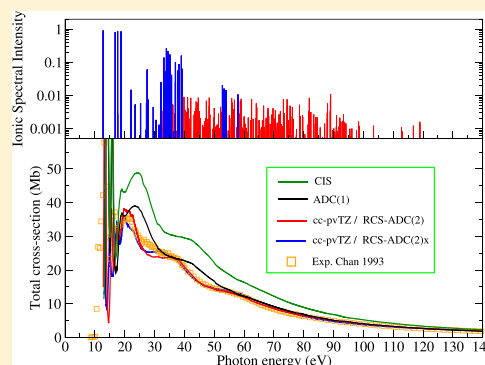


Restricted Correlation Space B-Spline ADC Approach to Molecular Ionization: Theory and Applications to Total Photoionization Cross-Sections

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ABSTRACT: Herein is presented a new approach to the ab initio algebraic diagrammatic construction (ADC) schemes for the polarization propagator, which is explicitly designed to accurately and efficiently describe molecular ionization. The restricted correlation space (RCS) version of the ADC methods up to second order of perturbation theory is derived via the intermediate state representation (ISR) and implemented in the multicenter B-spline basis set for the electronic continuum. Remarkably a general close-coupling structure of the RCS-ADC many-electron wave function, connecting the N -particle to the $(N - 1)$ -particle ADC intermediate states, emerges naturally as a nontrivial result of the RCS ansatz. Moreover, the introduced RCS-ADC schemes prove to be particularly manageable from a computational point of view, overcoming the practical limitations of the conventional ADC approaches. The quality of the new RCS-ADC(n) approaches is verified by performing a series of total photoionization cross-section calculations on a test set of molecules. The excellent agreement of the results with existing accurate benchmarks demonstrates that the RCS versions of the ADC schemes are optimal and quantitatively accurate methods for studying multichannel molecular photoionization.



INTRODUCTION

Over the past few decades, the groundbreaking development of ultrashort laser pulses¹ in a broad range of the electromagnetic spectrum, from near-infrared femtosecond pulses to extreme ultraviolet (XUV) and X-ray attosecond ones, has enabled the experimental study of electron dynamics in atoms and molecules on its natural time scale,^{2–8} giving one a direct insight into how electronic rearrangement may affect chemical properties.^{9–12} Considerable effort has been made to produce controlled intense few-cycle XUV and X-ray laser pulses, using both table-top high harmonic generation (HHG)^{13–15} and free-electron laser (FEL) sources.^{16,17} In particular, the spectacular experimental progress in incorporating attosecond technology in FEL,^{18–20} in order to generate intense ultrashort XUV pulses with high spatial and temporal coherence,²¹ and extending the HHG technology to reach the water window (3–4 nm wavelength),^{22–24} has opened the way to attosecond pump–probe experiments^{10,11,14,25–30} and the study of ultrafast correlated many-electron dynamics both in the valence and core energy regions.

The main fundamental physical process which takes place during the interaction of the molecular system with the experimentally available laser pulses is photoionization. Although valence and inner-valence ionization of molecular species with XUV light can eventually lead to more complex processes, such as multiple ionization by Auger cascades following inner-shell ionization and dissociative ionization, with multiply ionized molecular fragments as dominant final states of the system, the single-ionization process dominates as

a primary event. Moreover, multiple ionization processes tend to take place through a sequence of single-ionization steps. Therefore, an accurate theoretical description of the many-body physics underlying the photoionization process is essential in order to understand the ultrafast phenomena induced in atomic and molecular systems by the energetic electromagnetic radiation and to provide the necessary support that the sophisticated attosecond experiments require. Independently of whether the single-ionization event involves the exchange of one or several photons with the ionizing field, possibly with different frequencies, the photoionization process is described by the transition of the molecular many-electron system from a bound state to a continuum (scattering) state. Correlation effects can play a major role in the photoionization process, potentially giving rise to a series of many-electron phenomena such as shakeup effects, where the photon energy is shared between two electrons leading to the ionization of one and excitation of the other, the presence of satellite peaks associated with orbital relaxation,^{31–35} and breakdown of the molecular orbital (MO) picture, and nonradiative (auto-ionization and Auger) decay,^{36–38} associated with the formation of transiently bound, often multiply excited configurations,^{39,40} whose decay is due to the coupling between different ionization channels (configuration interaction (CI) in the continuum⁴¹). This calls for the development of first-principles theoretical tools capable of modeling, at

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the same level of accuracy, both the multiconfigurational nature of bound target states and the correlated dynamics in the ionization electronic continuum.

The basic computational problem one faces when trying to calculate many-electron wave functions belonging to the continuum part of the spectrum is how to take into account, with high accuracy, both the scattering character and long-range nature of the photoionized state wave function and the intricate correlated short-range structure of complex molecular systems. While the description of the ionization continuum is quite straightforward for atomic systems and many well-developed theoretical techniques exist for the description of atomic photoionization (see, e.g., refs 42–47 and references therein), the multicenter molecular problem still poses a formidable challenge to the theory, especially when increasing the size of the system. This is the reason why most theoretical studies on small and medium size molecules are still limited to a one-electron description of the phenomenon (single active electron (SAE) approximation methods)^{46,48–65} and partially disregard, in one way or another, electron correlation,¹² due to their inherent incapability of describing autoionization and Auger decay, as well as the coupling between continuum states associated with different states of the parent ion (interchannel coupling in the continuum).

On the other hand, although existing post-HF methods of ab initio quantum chemistry allow one to routinely obtain highly accurate many-electron wave functions and transition matrix elements for ground and excited bound states,^{66–69} the primitive finite sets of Gaussian-type orbitals (GTO) single-electron basis functions normally employed by these methods are of very limited use for the description of continuum states. Indeed, standard GTO basis functions (as well as Slater-type basis functions) are characterized by their exponential decrease that does not make them suitable to describe the oscillatory behavior of the continuum wave function up to large distances from the parent ion. Recent studies, performed by using the highly correlated algebraic diagrammatic construction (ADC)⁷⁰ and linear response coupled cluster methods⁷¹ for electronic excitations in conjunction with the Stieltjes–Chebyshev moment-theory technique,⁷² indicated that the use of GTOs in molecular photoionization cross-section calculations leads to the onset of major inaccuracies in the calculated cross-sections at about 70 eV above ionization threshold. Moreover, these studies showed that even very careful GTO selections cannot afford high-energy features and high resolution, without running into basis-set linear dependence problems.

In order to overcome the aforementioned limitations arising from the use of GTO basis sets, considerable effort has recently been made by several groups to combine the existing capability offered by quantum chemistry packages such as MOLCAS⁷³ and MOLPRO⁷⁴ with well-established techniques for the description of the electronic continuum. The common feature of these recently proposed methods is that they complement the short-range part of the wave function, represented by Gaussian functions, with other functions more suited to describe the unbound photoelectron, such as finite-element (FE) representation of the radial coordinate,⁷⁵ discrete variable representation (DVR),^{76–78} B-splines,⁷⁹ and even plane waves.⁸⁰ The close-coupling (CC) expansion method recently presented by Scrinzi et al.,⁵ based on a quantum chemistry multireference configuration interaction (MRCI) description of initial and target states, has been applied with success to the

strong-field ionization of atoms and small molecules.^{75,81,82}

The group of Martin et al. also developed a CC-expansion based approach (XCHEM⁸³), which is based on a quantum chemistry restricted-active-space self-consistent-field (RASSCF) description of initial and target states and has been applied with success to the calculation of total and partial photoionization cross-sections of atoms^{83,84} and molecular nitrogen.^{85,86} Furthermore, a single-center B-spline implementation of the configuration interaction singles (CIS) method has recently been proposed by Toffoli and Decleva⁸⁷ and applied to the calculation of atomic and molecular photoionization cross-sections.

Another very successful approach to describe atomic and molecular ionization, the ab initio B-spline ADC method,^{88–90} has been developed in the past few years within a collaboration between the group of Averbukh and Ruberti and the group of Decleva. The ab initio B-spline ADC method combines the capability of state-of-the-art many-body Green's function ADC techniques for the calculation of correlated excited states^{91–95} with an accurate description of the electronic continuum given by the B-spline basis. The B-spline implementation of the excitation ADC schemes explicitly treating not only single [ADC(1)] but also double [ADC(2)] excitations, for neutral atomic systems, has been first introduced for calculation of photoionization cross-sections of closed-shell atoms.⁸⁸ The time-dependent (TD) version of the atomic B-spline ADC(1) method has also been successfully applied to describe HHG spectra⁸⁸ and transient-absorption dynamics in closed-shell atoms.²⁷ Recently, extension of the TD B-spline ADC method to describe neutral and ionic molecular systems has been presented and successfully applied to the study of molecular HHG electron dynamics⁸⁹ and to the fully ab initio simulation of a prototypical molecular pump–probe experiment in the CO₂ molecule.⁹⁰ In these works, the neutral molecular system was described by using the lowest (first-) order of the ADC(*n*) hierarchy, i.e., ADC(1). Moreover, the employed single-electron basis was monocentric and consisted of the spherical harmonics for the angular part and a B-spline expansion for the radial coordinate.

The extension of the standard single-center B-spline ADC method to both larger polyatomic molecules and to higher orders of the ADC(*n*) hierarchy (i.e., ADC(2), ADC(2)*x*), in order to describe electron-correlation effects present when ionizing molecules from the inner-valence energy region, such as shake up, is extremely demanding from the computational point of view and can become rather impractical for a number of reasons. First the single-center expansion of the wave function is not well suited to describe molecular orbitals localized away from the center of the expansion, i.e., core orbitals, and suffers from convergence issues. Second, the number of basis functions used in B-spline ADC is well above the standards of a typical ab initio calculation. Because of this, and the fact that molecular symmetry is in general much lower with respect to the atomic case, the Hamiltonian matrix dimensions for molecules, represented using high-quality single-electron basis sets for the electronic continuum, rapidly become extremely, almost impracticably, large, easily exceeding the millions in computational schemes going beyond single excitations, e.g., ADC(2) and ADC(2)*x*. Moreover, the extremely large number (*m*) of basis functions makes the evaluation of the virtual summations present in the ADC formulas and coming from the perturbative expansion of the ground state extremely time-consuming due to the unfavorable

scaling (e.g., the scaling in ADC(2) is respectively m^4 and m^5 for the generation of the 1h1p block of the Hamiltonian and dipole matrices). Last but not least, the ADC(n) schemes, like other common quantum chemistry approaches to the many-body problem, do not lend themselves naturally to the disentanglement of the N -electron wave function structure, across a continuous range of energies, into the asymptotic wave function of the ejected electron and the states of the parent molecular ion. This is usually achieved in CC methods by augmenting the parent-ion states with electrons distributed in a large set of orbitals capable of reproducing the periodic oscillations characteristic of asymptotically free states, at the expense of encountering nontrivial overcompleteness problems⁹⁶ that require special treatment.

In this work I present a new and efficient formulation of the ab initio ADC(n) schemes that overcomes all the aforementioned limitations and is optimally designed for studying molecular ionization: B-spline restricted correlation space (RCS)-ADC. RCS-ADC is formulated using noncanonical HF orbitals and is based on the partition of the HF virtual space into two orthogonal subspaces. One space contains the orbitals used to describe the MP(n) correlation of the ground state (RCS), while the other orthogonally complements it to fully describe the ionization continuum and electrons reaching far out of the molecular region (ionization space (IS)). Moreover, within RCS-ADC, double excitations into the ionization space are excluded, keeping only 2h2p configurations in the RCS or mixed 2h2p configurations, where one electron is excited in the RCS and the other in the IS. This allows one to restrict the double-excitation space without losing the ability to accurately describe the single-ionization event (one electron in the continuum) and the correlation of the many-electron bound states intrinsic to the ADC(n) methods. Moreover, this procedure naturally gives rise to a general CC structure for the RCS-ADC many-electron wave function. Finally, the presented implementation of the RCS-ADC(n) schemes is based on the use of a multicenter B-spline basis set consisting of B-splines functions centered on both the various atomic locations and the molecular center of mass.

It is the purpose of this work to present the newly designed B-spline RCS-ADC(n) schemes for multichannel molecular photoionization and their implementation in a multicenter B-spline basis set, as well as to assess the quality, generality, and robustness of the newly presented approaches, by reporting its proof-of-concept time-independent application to the calculation of a series of total molecular photoionization cross-sections over a wide range of photon energies from the outer valence to 100 eV above the ionization threshold. This is an important step in order to check the performance of the B-spline RCS-ADC methods in describing bound-continuum transitions, before moving to electron-dynamic time-dependent calculations which will be reported in a second work to follow. To achieve this goal, in the present work, I use a test set of four molecules of first-row elements, for which both high-quality experimental results and a series of GTO-based standard ADC calculations are available in the literature. As in the case of the previously reported GTO-based molecular^{70,97} and B-spline atomic conventional ADC photoionization cross-section calculations,⁸⁸ here the total photoionization cross-section is calculated by using the Stieltjes-imaging (SI) moment-theory technique,⁷² which allows one to extract the correctly normalized oscillator-strength density in the electronic continuum starting from a general discretized spectrum.

The use of the same technique allows one to directly compare the accuracy of the results obtained with the RCS-ADC schemes with those of the conventional ADC ones, as well as to verify the accuracy of the multicenter B-spline basis-set implementation with respect to the GTO one. Moreover, I also verify the stability of the RCS-ADC results with respect to the type and size of correlation space used, providing yet another test of the accuracy of the method. Finally, I quantify the deviation of the RCS-ADC(1), RCS-ADC(2), and RCS-ADC(2) $\times$ cross-sections from the experimental ones by computing their energy-dependent relative discrepancies over the covered photon energy region.

This article is organized as follows. **Theory** presents the derivation and the relevant aspects of the RCS-ADC approach to molecular electronic excited states and ionization, and **Implementation** describes the computational implementation of the new method. The content presented in **Results** is devoted to testing the performance of RCS-ADC by using different levels of electronic correlation. Finally, **Conclusions**, including future perspectives, are given.

THEORY

Within the hierarchy of the conventional ISR-ADC ab initio schemes for systems with a closed-shell neutral ground state, the many-electron states are constructed starting from correlated excited states⁹⁴ defined as

$$|\Psi_I\rangle^N = \hat{C}_I^{N\dagger} |\Psi_0\rangle \quad (1)$$

where the operators $\hat{C}_I^{N\dagger}$ denote the physical excitation operators corresponding respectively to the different excitation classes of the neutral system ($I = 1p1h, 2p2h$, and so on),

$$\hat{C}_I^{N\dagger} = \{\hat{c}_a^\dagger \hat{c}_i; \hat{c}_a^\dagger \hat{c}_b^\dagger \hat{c}_j \hat{c}_k (a < b, j < k) \dots\} \quad (2)$$

and $|\Psi_0\rangle$ is the exact correlated ground state of the system. Here the subscripts $a, b, c, \dots$ and $i, j, k, \dots$ refer to the virtual and occupied HF orbitals, respectively.

Within the ADC approach, recursive orthonormalization of the excitation classes $|\Psi_I\rangle^N$ is performed by a two-step procedure consisting of a first Gram–Schmidt orthogonalization of the excitation class under consideration with respect to all the lower excitation classes followed by symmetric orthonormalization within the excitation class itself.^{94,95,98} This leads to the final expression for the ADC many-electron excited intermediate states $|\tilde{\Psi}_I\rangle^N$ in the form of a formal expansion in terms of the correlated ground state. The ADC secular matrix is the representation of the shifted electronic Hamiltonian $\hat{H} - E_0$ in the space of ADC intermediate states:⁹⁴

$$\mathcal{H}_{IJ} = \langle \tilde{\Psi}_I | \hat{H} - E_0 | \tilde{\Psi}_J \rangle \quad (3)$$

Within ADC, Møller–Plesset (MP) perturbation theory is introduced to describe the ground state correlation, i.e., $|\Psi_0\rangle$ and E_0 , and the hierarchy of ADC(n) approximations is obtained for each order n by truncating the excited state manifold at some limiting excitation class and, further, by consistently truncating the perturbation expansions for the included classes.^{91,92,94,98}

The core idea behind the RCS-ADC approach to molecular ionization is to separate the HF virtual space (VS) into two mutually orthogonal subspaces:

$$\{\phi_a\} = \{\chi_\alpha\} \oplus \{\psi_\mu\}, \quad \{\hat{c}_a^\dagger\} = \{\hat{c}_\alpha^\dagger\} \oplus \{\hat{c}_\mu^\dagger\} \quad (4)$$

where the subscripts $\alpha, \beta, \dots$ and $\mu, \nu, \dots$ refer to the two subspaces, respectively. Both sets of orthonormal spin-orbitals can be expressed as linear combinations of the canonical HF virtual orbitals,

$$\chi_\alpha = \sum_a C_a^\alpha \phi_a \quad \psi_\mu = \sum_a C_a^\mu \phi_a \quad (5)$$

but none of them diagonalizes the HF operator $\hat{F}$ on the entire virtual space. Indeed, the HF matrix in the transformed basis reads as

$$\mathbf{F} = \begin{pmatrix} \delta_{ij}\varepsilon_i & 0 & 0 \\ 0 & \delta_{\alpha\beta}\varepsilon_\alpha & F_{\alpha\mu} \\ 0 & F_{\mu\alpha} & \delta_{\mu\nu}\varepsilon_\mu \end{pmatrix} \quad (6)$$

While the HF operator is diagonal within each of the two subspaces, the HF coupling between the two is different from zero. Note that the single-particle basis-set transformation only affects the HF virtual space and that, consequently, the Brillouin theorem will still be valid within the present formulation of the RCS-ADC method; i.e., $F_{i\alpha} = F_{\mu i} = 0$. More specific details on how the two orthogonal subspaces are defined and calculated will be given in [Implementation](#).

The physical excitation operators can also be expressed in terms of the new single-particle virtual space basis as

$$\hat{C}_i^{N\dagger} = \{\hat{c}_\alpha^\dagger \hat{c}_i; \hat{c}_\mu^\dagger \hat{c}_i; \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_i \hat{c}_k (\alpha < \beta, j < k); \hat{c}_\mu^\dagger \hat{c}_\nu^\dagger \hat{c}_i \hat{c}_k (\mu < \nu, j < k); \hat{c}_\mu^\dagger \hat{c}_\alpha^\dagger \hat{c}_i \hat{c}_k \dots\} \quad (7)$$

Note that the new creation ($\hat{c}_\omega^\dagger$) and annihilation ($\hat{c}_\omega$) operators satisfy the following anticommutation relations:

$$\begin{aligned} \{\hat{c}_\alpha^\dagger, \hat{c}_\beta\} &= \delta_{\alpha\beta}; & \{\hat{c}_\alpha^\dagger, \hat{c}_\beta^\dagger\} &= \{\hat{c}_\alpha, \hat{c}_\beta\} = 0 \\ \{\hat{c}_\mu^\dagger, \hat{c}_\nu\} &= \delta_{\mu\nu}; & \{\hat{c}_\mu^\dagger, \hat{c}_\nu^\dagger\} &= \{\hat{c}_\mu, \hat{c}_\nu\} = 0 \\ \{\hat{c}_\alpha^\dagger, \hat{c}_\mu\} &= \{\hat{c}_\mu^\dagger, \hat{c}_\alpha\} = 0; & \{\hat{c}_\alpha^\dagger, \hat{c}_\mu^\dagger\} &= \{\hat{c}_\alpha, \hat{c}_\mu\} = 0 \end{aligned} \quad (8)$$

Given this factorization, the RCS ansatz for the correlated ground state $|\Psi_0\rangle$ reads as

$$\hat{c}_\mu |\Psi_0^{\text{RCS}}\rangle = 0 \quad \hat{c}_\alpha |\Psi_0^{\text{RCS}}\rangle \neq 0 \quad (9)$$

As a result of [eq 9](#), the RCS MP(n) perturbation expansion of the normalized correlated ground state $|\Psi_0\rangle$ reads (up to third order) as

$$\begin{aligned} |\Psi_0^{\text{RCS}}\rangle &= |\Phi_0^{\text{HF}}\rangle + |\Psi_0^{[1]\text{'RCS}}\rangle + |\Psi_0^{[2]\text{'RCS}}\rangle \\ &\quad - \frac{1}{2} |\Phi_0^{\text{HF}}\rangle \langle \Psi_0^{\text{RCS}} | \Psi_0^{\text{RCS}} \rangle^{[2]} + O[3] \\ |\Psi_0^{\text{'RCS}}\rangle &= |\Phi_0^{\text{HF}}\rangle + |\Psi_0^{[1]\text{'RCS}}\rangle + |\Psi_0^{[2]\text{'RCS}}\rangle + O[3] \end{aligned} \quad (10)$$

where the perturbation expansion, with respect to the zeroth-order HF ground state $|\Phi_0^{\text{HF}}\rangle$, is restricted to only one of the two HF-VS subspaces, i.e. $\{\chi_\alpha\}$. This subspace will be referred to as the restricted correlation space (RCS). The first order correction $|\Psi_0^{[1]\text{'RCS}}\rangle = \frac{1}{4} \sum_{\alpha\beta, ij} t_{ij}^{\alpha\beta} |\chi_\alpha \chi_\beta\rangle$, $\chi_\alpha \chi_\beta \in \text{RCS}$, contains only double excitations (2h2p) (with respect to the HF ground state $|\Phi_0^{\text{HF}}\rangle$), where the two excited electrons are promoted to RCS orbitals, while the second-order correction $|\Psi_0^{[2]\text{'RCS}}\rangle$ contains all excitations into the RCS from single (1h1p) to

quadruple (4h4p) ones. In these expressions, the quantity $t_{ij}^{\alpha\beta}$ stands for

$$t_{ij}^{\alpha\beta} = \frac{\langle \alpha\beta || ij \rangle}{\varepsilon_i + \varepsilon_k - \varepsilon_\alpha - \varepsilon_\beta} \quad (11)$$

and $\langle \alpha\beta || ik \rangle = \langle \alpha\beta | ik \rangle - \langle \alpha\beta | ki \rangle$ represents the antisymmetrized matrix element of the electron-electron Coulomb interaction, in terms of spin-orbitals $\chi(\sigma, \mathbf{r})$

$$\begin{aligned} \langle \alpha\beta || ik \rangle &= \int d\sigma_1 d\sigma_2 \int d^3\mathbf{r}_1 d^3\mathbf{r}_2 \chi_\alpha^*(\sigma_1, \mathbf{r}_1) \chi_\beta^*(\sigma_2, \mathbf{r}_2) \\ &\quad \times \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \chi_i(\sigma_1, \mathbf{r}_1) \chi_k(\sigma_2, \mathbf{r}_2) \end{aligned} \quad (12)$$

The corresponding ground state energy reads as, up to second order,

$$E_0^{[2]\text{RCS}} = E_0^{[0]} + \langle \Phi_0^{\text{HF}} | \hat{H} - \hat{F} | \Psi_0^{[1]\text{'RCS}} \rangle \quad (13)$$

The explicit expressions for the ADC excited states is obtained by performing the standard ISR-ADC two-step orthonormalization procedure and by using the excitation operators of [eqs 7](#) as well as the ground state expansion of [eq 10](#). The ADC excited states in the RCS maintain the same structure as in ordinary ISR-ADC. For example, the expansion of the first $|\tilde{\Psi}_i^{\alpha\beta}\rangle$ and second $|\tilde{\Psi}_{ij}^{\alpha\beta}\rangle$ excitation class states up to second and first order reads respectively as

$$\begin{aligned} |\tilde{\Psi}_i^{\alpha\beta}\rangle^{[2]} &= \hat{c}_\alpha^\dagger \hat{c}_i \{ |\Phi_0^{\text{HF}}\rangle + |\Psi_0^{(1)\text{'RCS}}\rangle + |\Psi_0^{(2)\text{'RCS}}\rangle \} \\ &\quad - |\Phi_0^{\text{HF}}\rangle \langle \Psi_0^{(2)\text{'RCS}} | \hat{c}_\alpha^\dagger \hat{c}_i | \Phi_0^{\text{HF}} \rangle \\ &\quad - \frac{1}{2} \sum_{\beta j} \hat{c}_\beta^\dagger \hat{c}_j \langle \Psi_0^{(1)\text{'RCS}} | \hat{c}_i^\dagger \hat{c}_\beta \hat{c}_\alpha^\dagger \hat{c}_i | \Psi_0^{(1)\text{'RCS}} \rangle + O[3] \end{aligned} \quad (14)$$

$$\begin{aligned} |\tilde{\Psi}_{ij}^{\alpha\beta}\rangle^{[1]} &= \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_i \hat{c}_j \{ |\Phi_0^{\text{HF}}\rangle + |\Psi_0^{(1)\text{'RCS}}\rangle \} \\ &\quad - |\Phi_0^{\text{HF}}\rangle \langle \Psi_0^{(1)\text{'RCS}} | \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_i \hat{c}_j | \Phi_0^{\text{HF}} \rangle + O[2] \end{aligned} \quad (15)$$

where the numbers in square brackets indicate the maximum perturbation order considered in the perturbative expansions. In what follows, I will denote the RCS intermediate states of the n th-order ADC(n) scheme as $|\tilde{\Psi}_{I_{\text{RCS}}}^N\rangle^{[n]}$.

Differently from the case of the excited intermediate states in the RCS, it turns out that, as shown below, the N -electron ADC excited states where one electron is promoted to orbitals belonging to the IS, can be expressed in terms of the $(N-1)$ -electron ionic ADC intermediate states, namely, in the form of so-called augmented states:

$$|\tilde{\Psi}_{\mu, I_{\text{RCS}}}^N\rangle = \hat{c}_\mu^\dagger |\tilde{\Psi}_{I_{\text{RCS}}}^{N-1}\rangle \quad (16)$$

where $\hat{c}_\mu^\dagger$ is the creation operator of an electron in the spin-orbital $\psi_\mu(\sigma, \mathbf{r})$ belonging to the IS and $|\tilde{\Psi}_{I_{\text{RCS}}}^{N-1}\rangle$ is a state of the (1h, 2h1p, and so on) ADC basis for the parent-ion in the RCS

$$|\tilde{\Psi}_{I_{\text{RCS}}}^{N-1}\rangle = \{ |\tilde{\Psi}_i^{N-1}\rangle; |\tilde{\Psi}_{aij}^{N-1}\rangle (i < j) \dots \} \quad (17)$$

In the following, the derivation of [eq 16](#) is outlined. Let us consider for example the first (1h1p) excitation class. The so-called precursor states read respectively as

$$|\Psi_{ak}^\# \rangle = \hat{c}_\alpha^\dagger \hat{c}_k |\Psi_0^{\text{RCS}}\rangle - |\Psi_0^{\text{RCS}}\rangle \langle \Psi_0^{\text{RCS}} | \hat{c}_\alpha^\dagger \hat{c}_k | \Psi_0^{\text{RCS}} \rangle \quad (18)$$

where

$$\rho_{ak}^{\text{RCS}} = \langle \Psi_0^{\text{RCS}} | \hat{c}_k^\dagger \hat{c}_\alpha | \Psi_0^{\text{RCS}} \rangle \quad (19)$$

is the RCS one-particle density matrix.

Using the RCS ansatz (see eq 9), it is immediate to see that, for IS excitation classes, $\rho_{\mu k} = 0$. As usual, the ISR-ADC intermediate states are obtained as

$$|\tilde{\Psi}_{ak}\rangle = \sum_{bl} |\Psi_{bl}^{\#}\rangle (\mathbf{S}^{-1/2})_{bl,ak} \quad (20)$$

where $\mathbf{S}$ is the overlap matrix of the first-excitation-class precursor states. Within RCS-ADC, the overlap matrix elements can be written as

$$S_{ak,\beta l} = \langle \Psi_0^{\text{RCS}} | \hat{c}_k^{\dagger} \hat{c}_a^{\dagger} \hat{c}_{\beta}^{\dagger} \hat{c}_l | \Psi_0^{\text{RCS}} \rangle = \rho_{ak}^{\text{RCS}} \rho_{\beta l}^{\text{RCS}} \quad (21)$$

$$S_{\mu k, \nu l} = \langle \Psi_0^{\text{RCS}} | \hat{c}_k^{\dagger} \hat{c}_{\mu}^{\dagger} \hat{c}_{\nu}^{\dagger} \hat{c}_l | \Psi_0^{\text{RCS}} \rangle = \delta_{\mu, \nu} \rho_{lk}^{\text{RCS}} \quad (22)$$

$$S_{ak, \nu l} = \langle \Psi_0^{\text{RCS}} | \hat{c}_k^{\dagger} \hat{c}_a^{\dagger} \hat{c}_{\nu}^{\dagger} \hat{c}_l | \Psi_0^{\text{RCS}} \rangle = 0 \quad (23)$$

where eqs 22 and 23 are obtained by using the anticommutation relations of eq 8. Therefore, given the ionic ADC intermediate states in the form $|\tilde{\Psi}_k^{N-1}\rangle = \sum_l \hat{c}_l |\Psi_0^{\text{RCS}}\rangle (\rho^{\text{RCS}})_{kl}^{-1/2}$ (see, for example, refs 98 and 99), eq 16 is finally obtained

$$\begin{aligned} |\tilde{\Psi}_{\mu, k}^N\rangle &= \sum_{\nu l} \hat{c}_{\nu}^{\dagger} \hat{c}_l |\Psi_0^{\text{RCS}}\rangle (\mathbf{S}^{-1/2})_{\nu l, \mu k} \\ &= \hat{c}_{\mu}^{\dagger} \sum_l \hat{c}_l |\Psi_0^{\text{RCS}}\rangle (\rho^{\text{RCS}})_{kl}^{-1/2} \\ &= \hat{c}_{\mu}^{\dagger} |\tilde{\Psi}_k^{N-1}\rangle \end{aligned} \quad (24)$$

The same procedure can be applied to higher excitation classes, therefore obtaining the general relation

$$|\tilde{\Psi}_{\mu, I_{\text{RCS}}}^N\rangle = \hat{c}_{\mu}^{\dagger} |\tilde{\Psi}_{I_{\text{RCS}}}^{N-1}\rangle \quad I_{\text{RCS}} = 1h, 2h1p, 3h2p, \text{ etc.} \quad (25)$$

where $|\tilde{\Psi}_{I_{\text{RCS}}}^{N-1}\rangle$ denotes the ISR RCS-ADC intermediate states of the $(N-1)$ -electron system (the derivation of eq 25 for the case of 2h1p ionic states is outlined in the Appendix). For example, the expansion of the IS first-excitation-class states up to second order reads as

$$\begin{aligned} |\tilde{\Psi}_{\mu, i}^N\rangle^{[2]} &= \hat{c}_{\mu}^{\dagger} \left\{ \hat{c}_i |\Phi_0^{\text{HF}}\rangle + \hat{c}_i |\Psi_0^{(1)'}\rangle + \hat{c}_i |\Psi_0^{(2)}\rangle \right. \\ &\quad \left. - \frac{1}{2} \sum_j \hat{c}_j |\Phi_0^{\text{HF}}\rangle \langle \Psi_0^{(1)'} | \hat{c}_j^{\dagger} \hat{c}_i | \Psi_0^{(1)'} \rangle \right\} \\ &= \hat{c}_{\mu}^{\dagger} |\tilde{\Psi}_i^{N-1}\rangle^{[2]} \end{aligned} \quad (26)$$

while the IS second-excitation-class states up to first order are given by

$$|\tilde{\Psi}_{\mu, aij}^N\rangle^{[1]} = \hat{c}_{\mu}^{\dagger} \{ \hat{c}_a^{\dagger} \hat{c}_i \hat{c}_j |\Phi_0^{\text{HF}}\rangle + \hat{c}_a^{\dagger} \hat{c}_i \hat{c}_j |\Psi_0^{(1)'}\rangle \} = \hat{c}_{\mu}^{\dagger} |\tilde{\Psi}_{aij}^{N-1}\rangle^{[1]} \quad (27)$$

where one can see the difference with respect to the standard form of the ADC excited states of eqs 14 and 15.

With an analogous derivation (see Appendix for the details), it is also possible to show that the ADC doubly excited states in the IS can be written as

$$|\tilde{\Psi}_{\mu\nu, I_{\text{RCS}}}^N\rangle = \hat{c}_{\mu}^{\dagger} \hat{c}_{\nu}^{\dagger} |\tilde{\Psi}_{I_{\text{RCS}}}^{N-2}\rangle \quad (28)$$

where $|\tilde{\Psi}_{I_{\text{RCS}}}^{N-2}\rangle$ is a state of the (2h, 3h1p, and so on) ADC basis for the doubly ionized $(N-2)$ -electron system in the RCS. The augmented state structure (see eqs 16 and 28) that links the RCS-ADC(n) N -electron intermediate excited states, where m electrons are promoted to the IS, to the respective ADC(n) intermediate states of the $(N-m)$ -electron system, is completely general and applies to every order n of the ADC hierarchy, reflecting the fact that the IS orbitals are orthogonal to the orbitals (RCS) describing the RCS ground state.

Note that the RCS intermediate states $|\tilde{\Psi}_{I_{\text{RCS}}}^N\rangle$ also include and therefore describe, although represented in a different way, the antisymmetrized products of RCS-ADC ionic states $|\tilde{\Psi}_{I_{\text{RCS}}}^{N-1}\rangle$ and RCS orbitals χ_w while the doubly excited configurations of the $\hat{c}_{\mu}^{\dagger} \hat{c}_{\nu}^{\dagger} |\tilde{\Psi}_{I_{\text{RCS}}}^{N-2}\rangle$ type further contribute to the description of bound excitations of the neutral system and doubly ionized states with two electrons in the continuum, as well as states representing single ionization leaving the parent ion in an IS 2h1p excited state, $|\tilde{\Psi}_{\nu, I_{\text{RCS}}}^{N-1}\rangle = \hat{c}_{\nu}^{\dagger} |\tilde{\Psi}_{I_{\text{RCS}}}^{N-2}\rangle$.

In this work, I restrict double excitations to a maximum of one electron promoted to unoccupied spin-orbitals belonging to the IS, therefore omitting the $|\tilde{\Psi}_{\mu\nu, I_{\text{RCS}}}^N\rangle$ states and including in the configuration space of the system only the $|\tilde{\Psi}_{I_{\text{RCS}}}^N\rangle$ and $|\tilde{\Psi}_{\mu, I_{\text{RCS}}}^N\rangle$ excited states. By doing so, I do not allow for the simultaneous excitation of two electrons in the continuum and I furthermore restrict the description of the parent ion to be within RCS-ADC ionic intermediate states only, without further contributions from orbitals/(2h1p excitations) in the IS.

In the conventional formulation of ADC the couplings between the ground state and all other intermediate states are, by definition, exactly zero at all orders, and consideration of the ground state, as well as of the spaces of uncoupled higher excitations, is therefore removed from the computation altogether. On the contrary, within RCS-ADC the IS excited configurations are coupled by the Hamiltonian to $|\Psi_0^{\text{RCS}}\rangle$. The couplings $\langle \Psi_0^{\text{RCS}} | (\hat{H} - E_0) \hat{c}_{\mu}^{\dagger} |\tilde{\Psi}_i^{N-1}\rangle$ and $\langle \Psi_0^{\text{RCS}} | (\hat{H} - E_0) \hat{c}_{\mu}^{\dagger} |\tilde{\Psi}_{aij}^{N-1}\rangle$, calculated respectively at second- and first-order of perturbation theory, read as

$$\begin{aligned} \langle \Psi_0^{\text{RCS}} | (\hat{H} - E_0) \hat{c}_{\mu}^{\dagger} |\tilde{\Psi}_i^{N-1}\rangle^{[2]} \\ = \sum_{\beta \in \text{RCS}} \rho_{\beta\mu}^{[2]} F_{\beta\mu} + \frac{1}{2} \sum_{\gamma \delta \in \text{RCS}, k} (t_{ik}^{\gamma\delta})^* \langle \gamma \delta || \mu k \rangle \end{aligned} \quad (29)$$

$$\langle \Psi_0^{\text{RCS}} | (\hat{H} - E_0) \hat{c}_{\mu}^{\dagger} |\tilde{\Psi}_{aij}^{N-1}\rangle^{[1]} = \langle ij || \mu a \rangle + \sum_{\beta \in \text{RCS}} (t_{ij}^{\beta a})^* F_{\beta\mu} \quad (30)$$

Diagonalization of the ionic ADC Hamiltonian in the RCS space

$$\mathbf{H}_{\text{RCS}}^{N-1} \mathbf{V}^{\text{ion}} = \mathbf{E}^{N-1} \mathbf{V}^{\text{ion}} \quad (31)$$

allows one to express the RCS-ADC ionic eigenstates (with given ionization energy $E_n^{N-1} - E_0^{\text{RCS}}$) in the basis of ionic RCS intermediate states as

$$|\Psi_n^{N-1}\rangle = \sum_i V_{i,n}^{\text{ion}} |\tilde{\Psi}_i^{N-1}\rangle + \sum_{aij} V_{aij,n}^{\text{ion}} |\tilde{\Psi}_{aij}^{N-1}\rangle \quad (32)$$

In turn, this transformation allows one to rotate the IS many-electron ADC configuration basis states $|\tilde{\Psi}_{\mu, I_{\text{RCS}}}^N\rangle$ and express them in terms of the RCS-ADC ionic eigenstates as

$$|\tilde{\Psi}_{\mu,n}^N\rangle = \hat{c}_{\mu}^{\dagger}|\Psi_n^{N-1}\rangle \quad (33)$$

Therefore, a generic ADC(n) many-electron wave function can be written as

$$|\Psi^N\rangle^{[n]} = \sum_m \left\{ \sum_{\mu} c_{m\mu} \hat{c}_{\mu}^{\dagger} |\Psi_m^{(N-1)}\rangle^{[n]} \right\} + \sum_{I_{RCS}} c_{I_{RCS}} |\tilde{\Psi}_{I_{RCS}}^N\rangle^{[n]} + c_0 |\Psi_0^{RCS}\rangle^{[n]} \quad (34)$$

where $|\Psi_m^{(N-1)}\rangle^{[n]}$ is a parent-ion state also consistently calculated at the ADC(n) level of theory. Hence, at the ADC(2) x level of theory, which is the highest in the ADC(n) hierarchy used in this work, the many-electron wave function reads as

$$|\Psi^N\rangle^{[2]x} = \sum_m \left\{ \sum_{\mu} c_{m\mu} \hat{c}_{\mu}^{\dagger} |\Psi_m^{(N-1)}\rangle^{[2]x} \right\} + \sum_{\alpha i} c_{\alpha i} |\tilde{\Psi}_i^{\alpha}\rangle^{[2]} + \sum_{\alpha < \beta, i < j} c_{\alpha\beta ij} |\tilde{\Psi}_{ij}^{\alpha\beta}\rangle^{[1]} + c_0 |\Psi_0^{RCS}\rangle^{[2]} \quad (35)$$

Spin adaptation of the primitive ADC basis functions can be obtained by applying to the whole set of RCS-ADC intermediate states the spin projection operator $\hat{P}_{S\Sigma}$ on the space of many-electron states with total spin S and spin projection Σ . Since in the present work photoionization from the closed-shell ground state is studied, here only singlet ($S = 0$) N -electron states are used.

While the basis set of the entire virtual space is designed to describe both bound states and the ionization continuum, within RCS-ADC both the bound ionic states and the GS are described exclusively by the RCS single-particle basis set. Moreover, the long-range part of the photoionized electron wave function should be exclusively described by the complementary IS single-particle basis set. This allows one to restrict the double-excitation space without losing the ability to accurately describe the single-ionization event (one electron in the continuum) and to consequently obtain a close-coupling structure for the many-electron wave function. Since exchange and correlation effects with the bound electrons are short-range, it is essential that the RCS basis set be designed to accurately describe the localized bound correlation of the system, while the IS space complements this basis in order to describe the photoelectron wave function over the entire space including the exclusive description of the long-range part of the photoelectron wave function.

Such basis sets have been extensively studied for many years within the quantum chemistry community and successfully optimized in order to describe the properties of molecular bound states. Therefore, I take advantage of this valuable resource and use as basis set for the RCS the optimized GTO basis sets of common use in the QC community, such as the Dunning's correlation consistent (cc-pVXZ) basis sets which have already been optimized using correlated (CISD) many-electron wave functions. The virtual orbitals obtained from an HF calculation in one of these GTO bases, such as cc-pVTZ, are projected onto the space of HF virtual orbitals computed using the entire single-particle basis set. The occupied HF orbitals, calculated by solving the HF equations over the full single-particle space, exhibit a minimal change with respect to the ones obtained from a GTO HF calculation. As a consequence, the GTO-based virtual space is a subspace of the total virtual space and therefore the nature of the GTO-expanded virtual orbitals is not altered by the aforementioned

projection operation. Finally, the IS space is obtained as the orthogonal complement, over the total (full-basis) virtual space, to the RCS.

Each of the terms in the wave function expansion (eq 34) can describe bound excitation of the neutral system, including the bound part of resonance states in the continuum, such as doubly excited states, that arise naturally from the CC expansion if one of them plays a key role in the dynamics. On the other hand, the description of the long-range scattering part of the wave function is given exclusively by the mutually orthogonal IS augmented states, which allows for the clear identification, in the N -electron wave function expansion, of the relevant ionic states participating in the ionization dynamics and takes into account the fact that when separated by large distances, the parent ion and the emitted electron do not interact. The general CC structure of the RCS-ADC wave function, where the IS configuration space is expressed in terms of a linear combination of antisymmetrized products of states of the parent ion and states of the asymptotically free electron, is an immediate consequence of the perfect factorization of the intermediate states into the ion times continuum form (eq 16) and of the rotation of the intermediate state basis (eqs 32 and 33). Note that all of the states in the N -electron wave function expansion are orthogonal to each other by construction. Moreover the orthogonality between the ionic target states (described by RCS orbitals) and the IS orbitals (which describe the photoelectron wave function outside of the bound and localized RCS space) was enforced when constructing the two RCS and IS orbital subspaces. A significant advantage with respect to standard CC formulations is that, in principle, all of the ionic eigenstates as obtained from an ionic-ADC calculation within the RCS single-particle basis set are automatically included in the CC expansion (eq 34) and therefore the theoretical description of the parent ion arises naturally from the ADC wave function expansion in the ADC configurations basis, without the need for an ad hoc inclusion. The same applies to the RCS-ADC correlated short-range states. Moreover, the expansion of eq 34 does not give rise to the usual overcompleteness issues of standard CC expansions, as the orthogonality of the IS orbitals, with respect to the orbitals describing the target states (RCS), is automatically taken care of by eq 4.

The flexibility of the RCS basis allows one to achieve a description of the correlation intrinsic to the ADC(n) methods, both in the MP(n) ground state and the many-electron ionic states, as accurate as in the calculations of standard QC literature, including the ability to describe hole-mixing, dominant-satellite, and breakdown of the molecular orbital picture scenarios. This formulation also opens up the possibility of, depending on the specific physical process under investigation, selecting out of all of the RCS-ADC ionic states the open ionic channels to be included in the calculation (i.e., in the CC expansion of eq 34). As a result, the ionic states excluded from the CC expansion are described exclusively as closed ionic channels, i.e., describing excited resonance states, by the $|\tilde{\Psi}_{I_{RCS}}^N\rangle$ terms. The structure of eq 34, which links the RCS-ADC(n) N -electron wave function to the respective ADC(n) ($N - 1$)-electron eigenstates, is general and applies to every order n of the ADC hierarchy. As a result, within the RCS-ADC(n) schemes for molecular ionization, electron correlation is included in the ab initio description of

multichannel molecular ionization at the systematically improvable level.

At the ADC(n) level of theory, the total RCS-ADC Hamiltonian matrix on the transformed basis contains the standard ADC(n) Hamiltonian within the RCS excitation space augmented with the $\hat{c}_\mu^\dagger |\Psi_n^{N-1}\rangle$ product states block, which describes ionization and interaction in the continuum, and reads as

$$\mathbf{H}^{\text{RCS-ADC}(n)} = \begin{pmatrix} \mathbf{H}_{\text{RCS}}^{\text{ADC}(n)} & \langle \tilde{\Psi}_{I_{\text{RCS}}}^N | (\hat{H} - E_0^{\text{RCS}}) \hat{c}_\mu^\dagger | \Psi_n^{N-1} \rangle^{\text{ADC}(n)} \\ \langle \Psi_m^{N-1} | \hat{c}_\mu (\hat{H} - E_0^{\text{RCS}}) | \tilde{\Psi}_{I_{\text{RCS}}}^N \rangle^{\text{ADC}(n)} & \langle \Psi_m^{N-1} | \hat{c}_\mu (\hat{H} - E_0^{\text{RCS}}) \hat{c}_\nu^\dagger | \Psi_n^{N-1} \rangle^{\text{ADC}(n)} \end{pmatrix} \quad (36)$$

For example, at the ADC(2) x level of theory, the rotated Hamiltonian is calculated by using the ADC(2) x ionic eigenstates $\mathbf{V}_m^{\text{ADC}(2)x}$ and by truncating both the excitation classes and the perturbation expansion of the matrix elements corresponding to the different excitation classes in a way consistent with the standard polarization propagator results:

$$\begin{aligned} \mathcal{H}_{\mu m, \mu n}^{\text{ADC}(2)x} &= \langle \Psi_m^{N-1} | \hat{c}_\mu (\hat{H} - E_0^{\text{RCS}}) \hat{c}_\nu^\dagger | \Psi_n^{N-1} \rangle^{\text{ADC}(2)x} \\ &= \sum_i \sum_k V_{k,m}^{\text{ADC}(2)x} V_{i,n}^{\text{ADC}(2)x} \langle \tilde{\Psi}_k^{N-1} | \hat{c}_\mu (\hat{H} - E_0) \hat{c}_\nu^\dagger | \tilde{\Psi}_i^{N-1} \rangle^{[2]} \\ &\quad + \sum_i \sum_{\beta k l} V_{\beta k l, m}^{\text{ADC}(2)x} V_{i, n}^{\text{ADC}(2)x} \langle \tilde{\Psi}_{\beta k l}^{N-1} | \hat{c}_\mu (\hat{H} - E_0) \hat{c}_\nu^\dagger | \tilde{\Psi}_i^{N-1} \rangle^{[1]} \\ &\quad + \sum_{a i j} \sum_k V_{k, m}^{\text{ADC}(2)x} V_{a i j, n}^{\text{ADC}(2)x} \langle \tilde{\Psi}_k^{N-1} | \hat{c}_\mu (\hat{H} - E_0) \hat{c}_\nu^\dagger | \tilde{\Psi}_{a i j}^{N-1} \rangle^{[1]} \\ &\quad + \sum_{a i j} \sum_{\beta k l} V_{\beta k l, m}^{\text{ADC}(2)x} V_{a i j, n}^{\text{ADC}(2)x} \langle \tilde{\Psi}_{\beta k l}^{N-1} | \hat{c}_\mu (\hat{H} - E_0) \hat{c}_\nu^\dagger | \tilde{\Psi}_{a i j}^{N-1} \rangle^{[1]} \end{aligned} \quad (37)$$

Furthermore, it is possible to show that the RCS-ADC(2) x Hamiltonian in the space of IS excited configurations, eq 37, can be written as the sum of two terms: one consisting of the sum of the second-order ionic ADC Hamiltonian and the zeroth-order IS orbital energy ϵ_μ and the other describing, at first order of perturbation theory, the ion-photoelectron intra- and interchannel couplings:

$$\begin{aligned} &\langle \tilde{\Psi}_{I_{\text{RCS}}}^{N-1} | \hat{c}_\mu (\hat{H} - E_0^{\text{RCS}}) \hat{c}_\nu^\dagger | \tilde{\Psi}_{J_{\text{RCS}}}^{N-1} \rangle^{[2]} \\ &= \langle \Phi_{\mu, I_{\text{RCS}}} | (\hat{H} - E_0^{\text{RCS}}) | \Phi_{\nu, J_{\text{RCS}}} \rangle^{\text{coup}} \\ &\quad + \delta_{\mu\nu} \{ \langle \tilde{\Psi}_{I_{\text{RCS}}}^{N-1} | (\hat{H} - E_0^{\text{RCS}}) | \tilde{\Psi}_{J_{\text{RCS}}}^{N-1} \rangle^{[2]} + \epsilon_\mu \} \end{aligned} \quad (38)$$

where $|\Phi_{\nu, I_{\text{RCS}}}\rangle$ are the zeroth-order configuration interaction excited states (1h1p, 2h2p) built on top of the HF ground state $|\Phi_0^{\text{HF}}\rangle$, and again the $I_{\text{RCS}}, J_{\text{RCS}} = k, \beta k l$ indexes are used for the 1h and 2h1p ionic configurations. Equation 38 can be readily obtained by considering all the Hamiltonian terms of eq 37 and collecting the ones that do not explicitly affect the IS-electron degree of freedom (i.e., the ones containing a $\delta_{\mu\nu}$ factor). The remaining terms are the first-order ion-photoelectron coupling terms and read respectively as

$$H_{\mu i, \nu j}^{[1], \text{coup}} = \langle \Phi_{\mu, i} | (\hat{H} - E_0^{\text{RCS}}) | \Phi_{\nu, j} \rangle^{\text{coup}} = -\langle \mu j || \nu i \rangle \quad (39)$$

$$\begin{aligned} H_{\mu i, \nu d k l}^{[1], \text{coup}} &= \langle \Phi_{\mu, i} | (\hat{H} - E_0^{\text{RCS}}) | \Phi_{\nu, d k l} \rangle^{\text{coup}} \\ &= \delta_{ik} \langle \mu l || \nu d \rangle - \delta_{il} \langle \mu k || \nu d \rangle \end{aligned} \quad (40)$$

$$\begin{aligned} H_{\mu b i j, \nu d k l}^{[1], \text{coup}} &= \langle \Phi_{\mu, b i j} | (\hat{H} - E_0^{\text{RCS}}) | \Phi_{\nu, d k l} \rangle^{\text{coup}} \\ &= \delta_{ik} \delta_{jl} \langle \nu d || \mu b \rangle - \delta_{jl} \delta_{bd} \langle i \nu || k \mu \rangle - \delta_{ik} \delta_{bd} \langle j \nu || l \mu \rangle \\ &\quad + \delta_{jk} \delta_{bd} \langle i \nu || l \mu \rangle + \delta_{il} \delta_{bd} \langle j \nu || k \mu \rangle \end{aligned} \quad (41)$$

Therefore, the following final expression is obtained:

$$\begin{aligned} &\langle \Psi_m^{N-1} | \hat{c}_\mu (\hat{H} - E_0^{\text{RCS}}) \hat{c}_\nu^\dagger | \Psi_n^{N-1} \rangle^{\text{ADC}(2)x} \\ &= \delta_{\mu\nu} \delta_{mn} \{ E_m^{(N-1), \text{ADC}(2)x} - E_0^{\text{RCS}} + \epsilon_\mu \} \\ &\quad + \sum_{I_{\text{RCS}}} \sum_{J_{\text{RCS}}} V_{I_{\text{RCS}}, m}^{\text{ADC}(2)x} V_{J_{\text{RCS}}, n}^{\text{ADC}(2)x} \langle \Phi_{\mu, I_{\text{RCS}}} | (\hat{H} - E_0^{\text{RCS}}) | \Phi_{\nu, J_{\text{RCS}}} \rangle^{\text{coup}} \end{aligned} \quad (42)$$

where the diagonal term describing the ADC(2) x ionization energies $E_m^{(N-1)} - E_0^{\text{RCS}}$ has been separated from the other (first-order) terms describing respectively the intrachannel potentials on the photoelectron and the interchannel couplings in the continuum. An analogous expression can be derived for the RCS-ADC(2) Hamiltonian. Note that, within RCS-ADC(2), the matrix elements between different 2h2p configurations in the expansion of eq 42 are neglected and only the contributions to the interchannel couplings coming from the interaction between different 1h states and between 1h and 2h1p states are considered. However, within RCS-ADC(2) x all the 2h2p–2h2p Hamiltonian couplings are evaluated at first order of perturbation theory. As a result, within RCS-ADC(2) x , interchannel couplings between different ionic states with strong 2h1p character are also described.

The off-diagonal blocks of eq 36, i.e., $\langle \tilde{\Psi}_{I_{\text{RCS}}}^N | (\hat{H} - E_0) \hat{c}_\nu^\dagger | \Psi_m^{N-1} \rangle^{\text{ADC}(n)}$, also contribute to describe the short-range interaction between the parent ion and the emitted electron, which is further dressed by the local ground state correlation built into each matrix element via perturbation theory. They are calculated in the same way as the previous block and the relevant matrix elements at the ADC(2) x level of theory read as follows:

$$\begin{aligned} H_{ai, \mu j}^{[2]} &= -\langle a j || \mu i \rangle - \frac{1}{4} \delta_{ij} \sum_{\gamma \in \text{RCS}, kl} [t_{kl}^{\alpha\gamma} \langle k l || \mu \gamma \rangle] \\ &\quad + \frac{1}{2} \sum_{\gamma \in \text{RCS}, k} [t_{ik}^{\alpha\gamma} \langle j k || \mu \gamma \rangle] + H_{ia, j\mu}^{(2)\text{HF}} + H_{ia, j\mu}^{(0)\text{HF}} \end{aligned} \quad (43)$$

where the last two terms

$$\begin{aligned} H_{ai, \mu j}^{(2)\text{HF}} &= -\frac{1}{4} \delta_{ij} \sum_{\gamma \delta \in \text{RCS}, kl} (t_{kl}^{\gamma\delta})^* t_{kl}^{\alpha\delta} F_{\gamma\mu} \\ &\quad + \frac{1}{2} \sum_{\gamma \delta \in \text{RCS}, k} (t_{kj}^{\gamma\delta})^* t_{ki}^{\alpha\delta} F_{\gamma\mu} \end{aligned} \quad (44)$$

$$H_{ai, \mu j}^{(0)} = \delta_{ij} F_{a\mu} \quad (45)$$

take into account the fact that neither the $\psi_\mu(\sigma, \mathbf{r})$ nor the $\chi_\alpha(\sigma, \mathbf{r})$ virtual spin-orbitals are eigenvectors of the HF operator on the complete basis set. As in the case of the conventional ADC(2) x schemes, the first-order matrix elements of the 1h1p–2h2p and 2h2p–2h2p blocks are given by $\langle \Phi_{a, I_{\text{RCS}}} | (\hat{H} - E_0^{\text{RCS}}) | \Phi_{\nu, J_{\text{RCS}}} \rangle$, $I_{\text{RCS}}, J_{\text{RCS}} = k, \beta k l$, while the 2h2p block also includes the contribution of a zeroth-order term

$$H_{a\beta ij, \gamma\mu kl}^{(0)\text{HF}} = \delta_{a\gamma} \delta_{ik} \delta_{jl} F_{\beta\mu} - \delta_{\beta\gamma} \delta_{ik} \delta_{jl} F_{a\mu} \quad (46)$$

Finally, let us consider the RCS-ADC representation of the one-particle dipole operator $\hat{D} = \sum_{rs} d_{rs} \hat{c}_r^\dagger \hat{c}_s$, where d_{rs} are the matrix elements of the dipole operator on the single-particle orbitals. As in the case of the Hamiltonian representation (eq 36), the transition matrix elements of the dipole operator between the RCS ground and excited states $\langle \Psi_0^{\text{RCS}} | \hat{D} | \tilde{\Psi}_{I_{\text{RCS}}}^N \rangle$ maintain the same structure as in the standard ADC(n) schemes,⁹² the only difference being that the virtual summations present in the formulas⁹² are restricted now to RCS orbitals only. On the contrary, the transition-dipole matrix elements between the RCS ground state and the IS excited states

$$\begin{aligned} \langle \Psi_0^{\text{RCS}} | \hat{D} \hat{c}_\mu^\dagger | \Psi_n^{N-1} \rangle &= \sum_{rs} d_{rs} \langle \Psi_n^{N-1} | \hat{c}_\mu \hat{c}_r^\dagger \hat{c}_s | \Psi_0^{\text{RCS}} \rangle \\ &= \sum_i V_{i,n}^{\text{ion}} \sum_{rs} d_{rs} \langle \tilde{\Psi}_i^{N-1} | \hat{c}_\mu \hat{c}_r^\dagger \hat{c}_s | \Psi_0^{\text{RCS}} \rangle \\ &\quad + \sum_{\alpha ij} V_{\alpha ij}^{\text{ion}} \sum_{rs} d_{rs} \langle \tilde{\Psi}_{\alpha ij}^{N-1} | \hat{c}_\mu \hat{c}_r^\dagger \hat{c}_s | \Psi_0^{\text{RCS}} \rangle \end{aligned} \quad (47)$$

simplify and can be written, using the RCS ansatz (eq 9) and the anticommutation relations of eq 8, in the following form:

$$\begin{aligned} \langle \Psi_0^{\text{RCS}} | \hat{D} \hat{c}_\mu^\dagger | \Psi_n^{N-1} \rangle &= \sum_I V_{I,n}^{\text{ion}} \sum_{\gamma \in \text{RCS}} d_{\gamma\mu} \langle \Psi_0^{\text{RCS}} | \hat{c}_\gamma^\dagger | \tilde{\Psi}_I^{N-1} \rangle \\ &= \sum_{\gamma \in \text{RCS}} d_{\gamma\mu} \langle \Psi_0^{\text{RCS}} | \hat{c}_\gamma^\dagger | \Psi_n^{N-1} \rangle \\ &= \sum_{\gamma \in \text{RCS}} d_{\gamma\mu} x_{\gamma n} \end{aligned} \quad (48)$$

where $x_{\gamma n}$ are the so-called spectroscopic amplitudes^{98,99}

$$x_{\gamma n} = \langle \Psi_0^{\text{RCS}} | \hat{c}_\gamma^\dagger | \Psi_n^{N-1} \rangle = \sum_I V_{I,n}^{\text{ion}} \langle \Psi_0^{\text{RCS}} | \hat{c}_\gamma^\dagger | \tilde{\Psi}_I^{N-1} \rangle \quad (49)$$

The spectroscopic amplitudes $x_{\gamma n}$ are related to the (relative) ionic photoelectron spectral intensities I_n , which can be approximately written as

$$I_n = \sum_\gamma |x_{\gamma n}|^2 \quad (50)$$

Note that, as a consequence of the strong-orthogonality condition $\langle \chi_\alpha | \psi_\mu \rangle = 0$ that holds between the RCS molecular orbitals describing the ionic target and the IS orbitals describing the outgoing electron, the transition matrix elements of eq 48 can also be equivalently expressed in terms of the Dyson orbital Δ_n associated with the ADC ionic eigenstate $|\Psi_n^{N-1}\rangle$

$$\langle \Psi_0^{\text{RCS}} | \hat{D} \hat{c}_\mu^\dagger | \Psi_n^{N-1} \rangle = \langle \Delta_n | \hat{d} | \psi_\mu \rangle \quad (51)$$

In order to compute the total photoionization cross-section of the N -electron system by using the Stieltjes-imaging method,⁷² it is necessary to calculate the spectrum of the system Hamiltonian and the transition moments from the initial bound state to the final unbound excited states.⁷⁰ This is done here by performing direct diagonalization of the ADC Hamiltonian (eq 36) in both the initial (ground) and final state spaces of the photoabsorption process. In turn, diagonalization of the ADC Hamiltonian

$$\begin{aligned} \mathbf{H}^N \mathbf{\Lambda}_0^N &= (E_0 - E_0^{\text{RCS}}) \mathbf{\Lambda}_0^N \\ \mathbf{H}^N \mathbf{\Lambda}_n^N &= (E_n^N - E_0^{\text{RCS}}) \mathbf{\Lambda}_n^N \end{aligned} \quad (52)$$

gives us the ground and excited eigenstates of the system in the full basis of intermediate states:

$$\begin{aligned} |\Psi_n^N\rangle &= \sum_{I_{\text{RCS}}} \Lambda_{I_{\text{RCS}},n}^N |\tilde{\Psi}_{I_{\text{RCS}}}^N\rangle + \sum_{\mu,m} \Lambda_{\mu m,n}^N \hat{c}_\mu^\dagger |\Psi_m^{N-1}\rangle \\ |\tilde{\Psi}_0^N\rangle &= \sum_{I_{\text{RCS}}} \Lambda_{I_{\text{RCS}},0}^N |\tilde{\Psi}_{I_{\text{RCS}}}^N\rangle + \sum_{\mu,m} \Lambda_{\mu m,0}^N \hat{c}_\mu^\dagger |\Psi_m^{N-1}\rangle \end{aligned} \quad (53)$$

while the vertical excitation energies are obtained as $E_n^N - E_0$. Having this explicit expression for the ground and excited states of the system, the transition moments of the type $\langle \tilde{\Psi}_0^N | \hat{D} | \Psi_n^N \rangle$ are given by

$$\langle \tilde{\Psi}_0^N | \hat{D} | \Psi_n^N \rangle = \mathbf{\Lambda}_0^{N\dagger} \mathbf{D} \mathbf{\Lambda}_n^N = \sum_{IJ} \Lambda_{I,0}^{N*} D_{IJ} \Lambda_{J,n}^N \quad (54)$$

Here D_{IJ} are the matrix elements of the dipole operator on the RCS-ADC many-electron basis-set functions, which at the ADC(2) x level of theory can be represented as

$$\mathbf{D}^{\text{RCS-ADC}(2)x} = \begin{pmatrix} \mathbf{D}_{\text{RCS}}^{\text{ADC}(2)x} & \langle \tilde{\Psi}_{\text{RCS}}^N | \hat{D} \hat{c}_\nu^\dagger | \Psi_m^{N-1} \rangle^{\text{ADC}(2)x} \\ \langle \Psi_m^{N-1} | \hat{c}_\mu \hat{D} | \tilde{\Psi}_{\text{RCS}}^N \rangle^{\text{ADC}(2)x} & \langle \Psi_m^{N-1} | \hat{c}_\mu \hat{D} \hat{c}_\nu^\dagger | \Psi_n^{N-1} \rangle^{\text{ADC}(2)x} \end{pmatrix} \quad (55)$$

where, as in the case of the RCS-ADC Hamiltonian matrix (eq 36), the blocks corresponding to $\hat{c}_\mu^\dagger |\Psi_n^{N-1}\rangle$ product states are calculated by using the ADC(2) x ionic eigenstates $\mathbf{V}_m^{\text{ADC}(2)x}$. In eq 55 the 1h1p–1h1p, 1h1p–2h2p, and 2h2p–2h2p matrix elements are evaluated respectively at second-, first-, and zeroth-order of perturbation theory.

As in the case of the $\langle \Psi_0^{\text{RCS}} | \hat{D} | \tilde{\Psi}_{\text{RCS}}^N \rangle$ transition dipoles, the matrix elements of the dipole operator between the RCS intermediate states $\langle \tilde{\Psi}_{I_{\text{RCS}}}^N | \hat{D} | \tilde{\Psi}_{J_{\text{RCS}}}^N \rangle$ have the same structure as in standard ADC(n),⁹⁴ the only difference being that the virtual summations present in the formulas⁹⁴ are restricted to RCS orbitals only. On the other hand, the form of the matrix elements between IS intermediate states $\langle \tilde{\Psi}_I^{N-1} | \hat{c}_\mu \hat{D} \hat{c}_\nu^\dagger | \tilde{\Psi}_J^{N-1} \rangle$ becomes considerably more simple and transparent. Indeed, using the anticommutation relations of eq 8 and the fact that, as a consequence of eq 9, $\hat{c}_\mu | \tilde{\Psi}_I^{N-1} \rangle = 0$, the following expression can be obtained:

$$\begin{aligned} \langle \tilde{\Psi}_I^{N-1} | \hat{c}_\mu \hat{D} \hat{c}_\nu^\dagger | \tilde{\Psi}_J^{N-1} \rangle &= \sum_{rs} d_{rs} \langle \tilde{\Psi}_I^{N-1} | \hat{c}_\mu \hat{c}_r^\dagger \hat{c}_s \hat{c}_\nu^\dagger | \tilde{\Psi}_J^{N-1} \rangle \\ &= d_{\mu\nu} \langle \tilde{\Psi}_I^{N-1} | \tilde{\Psi}_J^{N-1} \rangle + \delta_{\mu\nu} \sum_{rs} d_{rs} \langle \tilde{\Psi}_I^{N-1} | \hat{c}_r^\dagger \hat{c}_s | \tilde{\Psi}_J^{N-1} \rangle \\ &= \delta_{IJ} d_{\mu\nu} + \delta_{\mu\nu} \langle \tilde{\Psi}_I^{N-1} | \hat{D} | \tilde{\Psi}_J^{N-1} \rangle \end{aligned} \quad (56)$$

The off-diagonal blocks of the dipole matrix between RCS and IS ADC intermediate states, i.e., $\langle \tilde{\Psi}_{\text{RCS}}^N | \hat{D} \hat{c}_\nu^\dagger | \tilde{\Psi}_{\text{RCS}}^N \rangle$ and $\langle \tilde{\Psi}_{\text{RCS}}^{N-1} | \hat{c}_\nu \hat{D} | \tilde{\Psi}_{\text{RCS}}^N \rangle$, also considerably simplify with respect to the conventional formulation of ISR-ADC.⁹⁴ For completeness, the matrix elements that differ from the ones of the standard ISR-ADC formulation⁹⁴ are reported below. The second-order (RCS) 1h1p–(IS) 1h1p block matrix elements read as

$$\begin{aligned}
 \langle \tilde{\Psi}_{ai}^N | \hat{D} \hat{c}_\mu^\dagger | \tilde{\Psi}_j^{N-1} \rangle^{[2]} &= D_{ia,\mu}^{[2]} \\
 &= \delta_{ij} d_{a\mu} - \delta_{ij} \sum_l \rho_{al}^{[2]} d_{l\mu} - \frac{1}{4} \delta_{ij} \sum_{\gamma \delta \in \text{RCS}, kl} (t_{kl}^{\gamma\delta})^* t_{ik}^{\alpha\delta} d_{\gamma\mu} \\
 &\quad + \frac{1}{2} \sum_{\gamma \delta \in \text{RCS}, k} (t_{jk}^{\gamma\delta})^* t_{ik}^{\alpha\delta} d_{\gamma\mu}
 \end{aligned} \quad (57)$$

The first-order (IS)1h1p–(RCS)2h2p block matrix elements can be expressed as

$$\langle \tilde{\Psi}_i^{N-1} | \hat{c}_\mu^\dagger \hat{D} | \tilde{\Psi}_{ajk}^N \rangle^{[1]} = D_{i\mu, ajk}^{[1]} = \delta_{ij} \sum_l (t_{lk}^{\alpha\beta})^* d_{\mu l} - \delta_{ik} \sum_l (t_{lj}^{\alpha\beta})^* d_{\mu l}$$

while the form of the first-order (RCS)1h1p–(IS)2h2p block matrix elements is different and reads as

$$\langle \tilde{\Psi}_{ai}^N | \hat{D} \hat{c}_\mu^\dagger | \tilde{\Psi}_{bjk}^{N-1} \rangle^{[1]} = D_{ia, \mu bjk}^{[1]} = \delta_{a\beta} \delta_{ij} d_{k\mu} - \delta_{a\beta} \delta_{ik} d_{j\mu}$$

Finally the expression for the (RCS)2h2p–(IS)2h2p block matrix elements reads as

$$\langle \tilde{\Psi}_{\alpha\beta ij}^N | \hat{D} \hat{c}_\mu^\dagger | \tilde{\Psi}_{\gamma kl}^{N-1} \rangle^{[0]} = D_{\alpha\beta ij, \mu \gamma kl}^{(0)} = \delta_{\beta\gamma} \delta_{ik} \delta_{jl} d_{\alpha\mu} - \delta_{\alpha\gamma} \delta_{ik} \delta_{jl} d_{\beta\mu} \quad (60)$$

IMPLEMENTATION

In this work, the previous single-center B-spline implementation adopted within the standard ADC method has been extended to the more general multicenter approach based on B-spline functions. The multicenter LCAO B-spline basis, introduced by Toffoli and Decleva in the context of density functional theory, has already been described extensively in previous works.^{57,100–102} Therefore, here it suffices to provide a brief description of the multicenter B-spline basis set used.

The present implementation is limited to Abelian point groups that have real irreducible representations (irreps), namely, D_{2h} and its subgroups, and in the following, the nondegenerate irreps will be denoted by the index λ . The basis set employed in the present work consists of the following: (i) the large single-center B-spline basis-set expansion around the origin of coordinates (OCE), already adopted in the single-center B-spline ADC implementation,^{88–90}

$$\xi_{nlh\lambda} = \frac{1}{r} B_n(r) X_{lh\lambda}(\theta_{\text{OCE}}, \phi_{\text{OCE}}) X_{lh\lambda} = \sum_m b_{mnlh\lambda} Y_{lm} \quad (61)$$

and (ii) the OCE expansion supplemented with a smaller number of B-spline functions centered on the off-center positions of the nuclei j

$$\xi_{nlh\lambda}^i = \sum_{j \in Q_i} \frac{1}{r_j} B_n(r_j) \sum_m b_{mnlh\lambda j} Y_{lm}(\theta_j, \phi_j) \quad (62)$$

In eq 61, $X_{lh\lambda}(\theta_{\text{OCE}}, \phi_{\text{OCE}})$ are (LCAO) symmetry-adapted angular functions, which exploit molecular symmetry and are obtained from complex spherical harmonics Y_{lm} by a unitary transformation; h represents the number of linearly independent angular functions $X_{lh\lambda}$ for given l and symmetry λ ; m is the usual azimuthal angular momentum quantum number. The OCE expansion is completely defined by the B-splines order k , the radial grid (including the size of the radial interval $r_{\text{max}}^{\text{OCE}}$) and the maximum value of the angular momentum used in the expansion $l_{\text{max}}^{\text{OCE}}$ respectively. In eq 62, the index i runs over the non-equivalent (under symmetry operations) nuclei and Q_i is the set of equivalent nuclei, while the index j runs over the equivalent nuclei and gives the origin of the off-center spherical coordinates, r_j , θ_j , ϕ_j . The symmetry-adaptation coefficients

$b_{mnlh\lambda j}$ are again determined by the symmetry of the molecule. The off-center expansions, one for each set of equivalent centers, share the same B-spline order as the OCE expansion and are characterized by their linear radial grid (the size of the radial interval r_{max}^j and the step size h) and the maximum value of the angular momentum expansion l_{max}^j . In order to avoid running into linear dependency problems and consequent numerical instability, the spheres of radius r_{max}^j are not allowed to overlap.⁵⁷

Construction of the ADC Hamiltonian and dipole matrices requires carrying out restricted HF calculations and transforming the electron repulsion integrals from the multicenter primitive orbital basis to the basis of RCS and IS molecular orbitals (MOs). In order to construct the RCS and IS, I start from a set of occupied $\{\chi_i\}$, $i = 1, \dots, N/2$, and virtual $\{\tilde{\chi}_\alpha\}$, $\alpha = 1, \dots, N_g$, orthonormal MOs obtained as solutions of a GTO-based closed-shell restricted HF calculation performed with the MOLPRO⁷⁴ quantum chemistry program. Operatively, the set of MOs expanded in Gaussians are projected on the multicenter B-spline basis. The GTO-expanded occupied and virtual orbitals have HF energies $\{\tilde{\epsilon}_i\}$ and $\{\tilde{\epsilon}_\alpha\}$, respectively. However, the restricted closed-shell MOs obtained with the MOLPRO quantum chemistry program and projected on the multicenter basis are no longer SCF solutions of the HF Hamiltonian in the B-spline basis. Therefore, the HF Hamiltonian is rebuilt using these projected orbitals and iteratively diagonalized,

$$\hat{F}_{\text{GTO}} \phi_a = \epsilon_a \phi_a \quad (63)$$

to get a new set of $N/2$ orthonormal occupied orbitals $\{\chi_i\}$ as well as the complete set of orthonormal virtual orbitals in the B-spline basis set $\{\phi_a\}$, $a = 1, \dots, N_b$, $N_b > N_g$. The new occupied orbitals $\{\chi_i\}$ do not differ significantly from the ones obtained by MOLPRO $\{\tilde{\chi}_i\}$, assuming that an appropriate GTO basis has been used. Consequently, the GTO virtual space $\{\tilde{\chi}_\alpha\}$ is almost entirely contained in the total virtual space $\{\phi_a\}$. The use of a multicenter basis set dramatically improves the convergence of the approach with respect to the single-center implementation,^{57,88–90} especially in systems with heavy off-center nuclei, allowing one to converge the description of core orbitals with high accuracy, without running into linear dependency problems.

The RCS and its orthonormal complement IS are built by diagonalizing on the global virtual space $\{\phi_a\}$ the projector onto the space spanned by the fitted GTO virtual $\tilde{\chi}_\alpha$

$$\hat{P}^{\text{GTO}} = \sum_\alpha |\tilde{\chi}_\alpha\rangle \langle \tilde{\chi}_\alpha|; \quad \hat{P}^{\text{GTO}} \chi_\alpha = \chi_\alpha; \quad \hat{P}^{\text{GTO}} \psi_\mu = 0 \quad (64)$$

Finally, in order to simplify the ADC formulas, the HF operator is diagonalized onto both the RCS and IS spaces, giving

$$\hat{F} \chi_\alpha = \epsilon_\alpha \chi_\alpha, \quad \hat{F} \psi_\mu = \epsilon_\mu \psi_\mu \quad (65)$$

with the final RCS virtual orbital energies being approximately equal to the energies of the GTO-expanded virtual orbitals obtained from MOLPRO, i.e., $\epsilon_\alpha \approx \tilde{\epsilon}_\alpha$.

The two-electron integrals required within the RCS-ADC approach are of the four types

$$(\chi_{i,\alpha} \chi_{j,\beta} | \chi_{k,\gamma} \chi_{l,\delta}), \quad (\chi_{i,\alpha} \chi_{j,\beta} | \chi_{k,\gamma} \psi_\mu), \quad (\chi_{i,\alpha} \psi_\mu | \chi_{k,\beta} \psi_\nu), \quad (\chi_{i,\alpha} \chi_{j,\beta} | \psi_\mu \psi_\nu) \quad (66)$$

where only a maximum of one particle is described by IS orbitals and the RCS orbitals are MOs expanded in Gaussians and projected on the B-spline basis. Operatively, given the representation of the occupied $|\chi_i\rangle = \sum_\lambda \langle \xi_\lambda | \chi_i \rangle |\xi_\lambda\rangle$, RCS $|\chi_\alpha\rangle = \sum_\lambda \langle \xi_\lambda | \chi_\alpha \rangle |\xi_\lambda\rangle$, and IS $|\psi_\mu\rangle = \sum_\lambda \langle \xi_\lambda | \psi_\mu \rangle |\xi_\lambda\rangle$ MOs in the multicenter B-spline basis $\{\xi_\lambda\}$, the necessary integrals are obtained by transforming the two-electron integrals over the primitive LCAO B-spline functions,

$$(\chi_{i,\alpha} \xi_\mu | \chi_{k,\beta} \xi_\nu), \quad (\chi_{i,\alpha} \chi_{j,\beta} | \xi_\mu \xi_\nu) \quad (67)$$

into the RCS and IS MO basis set. The primitive basis two-electron integrals of eq 67 were obtained by using the numerical B-spline library developed by Toffoli and Decleva of the University of Trieste.¹⁰³

The two-electron integral storage requirement for each $\alpha\beta$ pair is N_b^2 , while the algorithm scaling with the number of pairs is of the order of N_g^2 . Note that, in the RCS-ADC(2) method, the two-electron integrals of the type $(\chi_\alpha \chi_\beta | \psi_\mu \psi_\nu)$ and $(\chi_\alpha \psi_\mu | \chi_\beta \psi_\nu)$ are not necessary, which makes the ADC(2) x method considerably more demanding from the computational point of view. One of the computational advantages of the RCS-ADC formulation is that all the summations over virtual orbital indexes in the perturbative ADC formulas are now restricted to the RCS χ_α virtual space orbitals, allowing for increasing the size of the B-spline basis without facing, when computing matrix elements involving continuum orbitals, the unfavorable N_b^4 scaling of the conventional ADC calculations.

In view of the large size of the B-spline bases (compared to GTO), optimized implementation of the B-spline RCS-ADC methods within the standard MPI and OpenMP protocols has been realized. Most importantly, this allows one to evenly distribute, across multiple processors, the memory storage associated with the different types of two-electron integrals required within the RCS-ADC methods. Finally, due to the reduced dimension of the full N -electron configuration space, that does not include double excitations where two electrons are simultaneously excited to IS orbitals; in this work it was possible to obtain the full spectrum of the ADC Hamiltonian matrices, required in the SI cross-section calculations, by performing direct full diagonalization of the ADC matrices.

RESULTS

The primary goal of this work is to test the accuracy of the newly designed and implemented B-spline RCS-ADC method at the ADC(2) and ADC(2) x levels of ab initio theory for bound-free transitions by calculating a series of molecular total photoionization cross-sections. For this purpose, I used a test set of molecules for which both accurate experimental results and a series of GTO-based calculations are available. As in the case of previous works reporting both GTO^{70,97} and atomic B-spline⁸⁸ conventional ADC calculations of total photoionization cross-sections, I use the Stieltjes-imaging (SI) technique,⁷² which allows for the extraction of the correctly normalized oscillator-strength density in the electronic continuum, starting from a general discretized spectrum. This enables one not only to directly compare the quality of the RCS-ADC methods for molecular ionization against the results obtained using the standard ADC formulation but also to verify the stability of the multicenter B-spline results with respect to the Stieltjes order, providing yet another test of the accuracy of the implementation. Stieltjes-imaging procedure is carried out in quadruple precision using the algorithm of ref 104. Moreover, I

restricted the calculated cross-sections to the energy range of up to 130 eV, thus including valence-type excitations but excluding excitations of the core electrons. Core-hole excitations, which can also be described by the RCS-ADC method, will be the object of future studies.

Here, B-spline functions of order 10 were used. Moreover, in this work, a linear grid of knots for the OCE expansion was supplemented with additional knots close to the nuclear positions, in order to improve the convergence of the radial expansion in eq 61. The range and step size h (distance between two consecutive knots) of the OCE grid used in this work are, for all the molecules considered in this work, $r_{\max}^{\text{OCE}} = 60$ au and $h = 0.8$ au, respectively. In the present work, photon energies up to ≈ 4 au are used. Therefore, the density of B-spline knots needs to be able to describe photoelectrons with wavelengths up to ≈ 2.2 au. In ref 105, a convergence study by Cormier et al. and Lambropoulos showed that a B-spline density (number of local B-splines per radial unit length) of 1.33 is sufficient for the convergence of the above-threshold ionization spectrum of hydrogen up to a photoelectron energy of 4.15 au. Therefore, in this work the B-spline grid I used is such that approximately 3 B-splines are contained in one wavelength of the most energetic photoelectrons. I found that this choice of parameters gives convergent results for the calculated cross-sections within the SI framework. All of the results presented here are converged with respect to the parameters of the calculation, namely, both the B-spline basis-set parameters, including the radial grid step, the range of the radial grid and the number of asymptotic angular momenta included, and the number of ADC ionic open channels included in the CC expansion of the many-electron wave function (see eq 35).

For every molecule studied in this work, it was found that the inclusion of the ionic channels characterized by a value of their spectral intensity (see eq 50) greater than 0.01 is a necessary as well as sufficient condition in order to achieve convergence of the calculated cross-sections, meaning that further inclusion, in the expansion of the many-electron wave function (eq 35) of ionic channels characterized by a smaller value of their spectral intensity did not modify the SI results for the cross-section. Experimental total photoionization cross-section as well as a series of SI results obtained via full diagonalization of the ADC(1), RCS-ADC(2), and RCS-ADC(2) x matrices are reported in Figures 1, 2, 3, and 4 for the molecules considered. In each figure, the spectral intensities of all of the ionic states calculated at the ADC(2) x level of theory as well as the ones included in the RCS-ADC cross-section calculations are also shown. In order to highlight the effect of electron correlation in the photoionization cross-sections, for each molecule, the cross-section was also calculated by using the CIS method, which does not include electron correlation.

For all of the systems considered here, the RCS was built using correlation consistent basis sets of the cc-pVTZ type, which have proven to be capable of yielding very accurate results when calculating the cross-sections by using second-order methods such as ADC(2) and ADC(2) x in combination with the SI technique. In order to investigate the quality of the cc-pVnZ-type basis sets in providing a good description of the RCS and the ionization cross-sections, in Figure 5 I compared the CO₂ total RCS-ADC(2) and RCS-ADC(2) x cross-sections calculated using respectively the cc-pVDZ RCS, energy-truncated cc-pVTZ RCS, and cc-pVTZ RCS. Finally, the deviation of the ADC(1), ADC(2), and ADC(2) x cross-sections from the experimental ones are quantified by

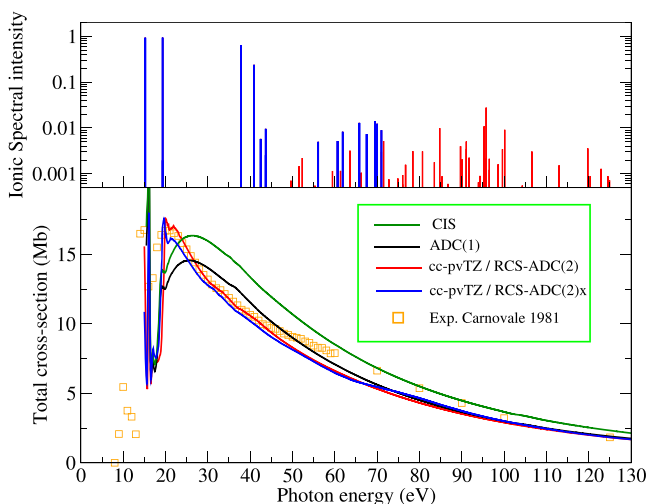


Figure 1. Top panel: Ionic cc-pVTZ ADC(2)x spectral intensities of the HF molecule (red); ionic states at which the convergence of the RCS-ADC(2)x total cross-section is achieved (blue). Bottom panel: Total photoionization cross-section of the HF molecule (squares, experimental results of ref 107; green line, B-spline CIS Stieltjes result; black line, B-spline ADC(1) Stieltjes result; red line, B-spline RCS-ADC(2) Stieltjes cross-section; blue line, B-spline RCS-ADC(2)x Stieltjes cross-section). The RCS-ADC(2) and RCS-ADC(2)x cross-sections are obtained using a RCS consisting of all the virtual orbitals from a GTO cc-pVTZ HF calculation.

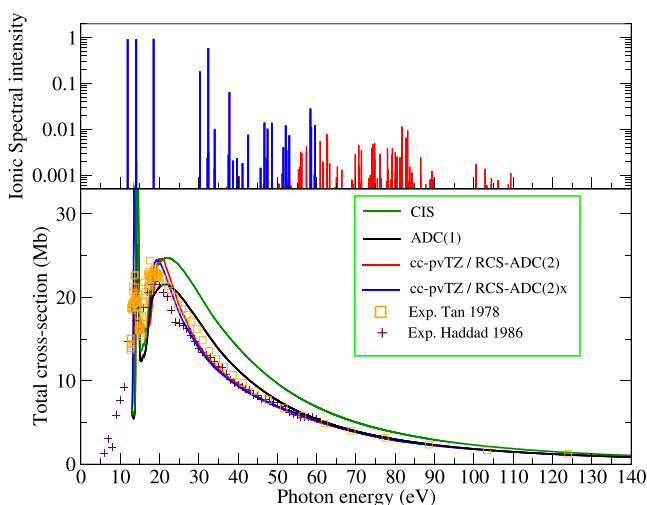


Figure 2. Top panel: Ionic cc-pVTZ ADC(2)x spectral intensities of the H₂O molecule (red); ionic states at which the convergence of the RCS-ADC(2)x total cross-section is achieved (blue). Bottom panel: Total photoionization cross-section of the H₂O molecule (squares, experimental results of ref 108; crosses, experimental result of ref 109; green line, B-spline CIS Stieltjes result; black line, B-spline ADC(1) Stieltjes result; red line, B-spline RCS-ADC(2) Stieltjes cross-section; blue line, B-spline RCS-ADC(2)x Stieltjes cross-section). The RCS-ADC(2) and RCS-ADC(2)x cross-sections are obtained using a RCS consisting of all the virtual orbitals from a GTO cc-pVTZ HF calculation.

computing their energy-dependent relative discrepancies over the covered photon energy region.

In the calculations presented here, the corrections to the ground state of the N -electron system coming from the Hamiltonian coupling terms of eqs 29 and 30 turn out to be negligible. Therefore, for the molecules and corresponding

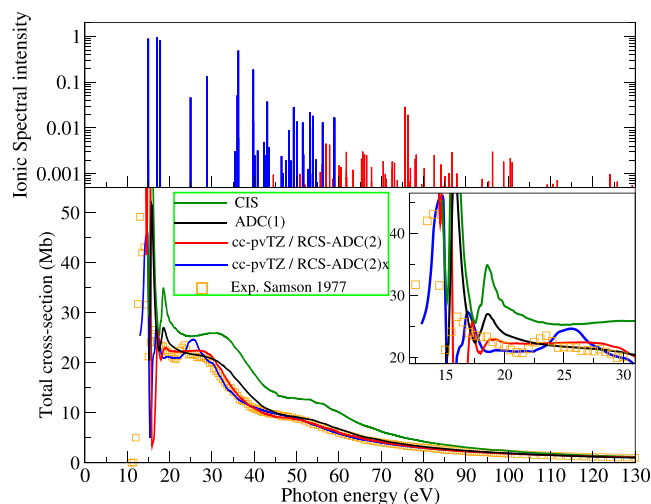


Figure 3. Top panel: Ionic cc-pVTZ ADC(2)x spectral intensities of the N₂ molecule (red); ionic states at which the convergence of the RCS-ADC(2)x total cross-section is achieved (blue). Bottom panel: Total photoionization cross-section of the N₂ molecule (squares, experimental results of ref 110; green line, B-spline CIS Stieltjes result; black line, B-spline ADC(1) Stieltjes result; red line, B-spline RCS-ADC(2) Stieltjes cross-section; blue line, B-spline RCS-ADC(2)x Stieltjes cross-section). The RCS-ADC(2) and RCS-ADC(2)x cross-sections are obtained using a RCS consisting of all the virtual orbitals from a GTO cc-pVTZ HF calculation. The inset better shows the results in the energy region between 18 and 30 eV.

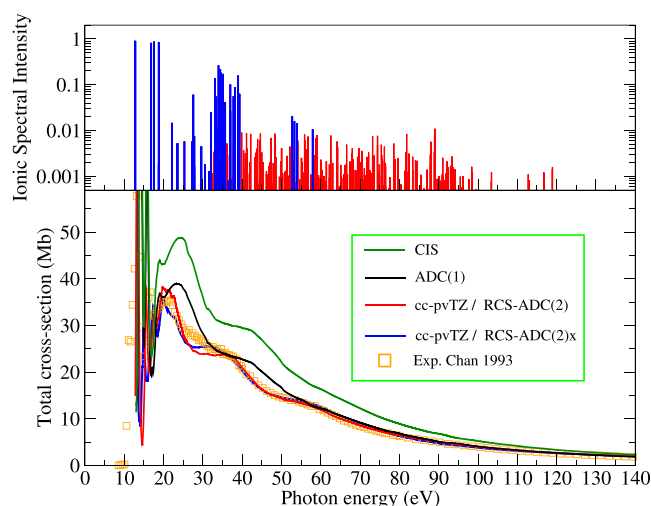


Figure 4. Top panel: Ionic cc-pVTZ ADC(2)x spectral intensities of the CO₂ molecule (red); ionic states at which the convergence of the RCS-ADC(2)x total cross-section is achieved (blue). Bottom panel: Total photoionization cross-section of the CO₂ molecule (squares, experimental results of ref 111; green line, B-spline CIS Stieltjes result; black line, B-spline ADC(1) Stieltjes result; red line, B-spline RCS-ADC(2) Stieltjes cross-section; blue line, B-spline RCS-ADC(2)x Stieltjes cross-section). The RCS-ADC(2) and RCS-ADC(2)x cross-sections are obtained using a RCS consisting of all the virtual orbitals from a GTO cc-pVTZ HF calculation.

RCS basis considered in this work, I found that the cross-sections calculated using the RCS $|\Psi_0^{\text{RCS}}\rangle$ or the corrected $|\Psi_0\rangle$ ground state of eq 53 are practically identical. However, note that in general this might not be the case when different systems are considered or when smaller GTO basis sets are used for the construction of the RCS.

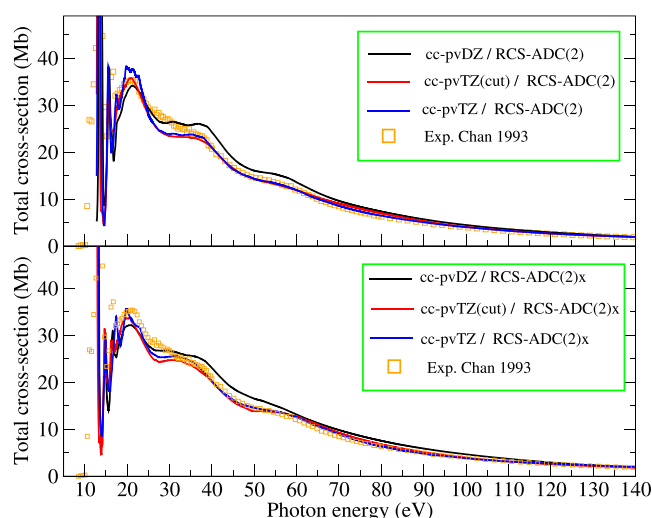


Figure 5. Top panel: B-spline RCS-ADC(2) total photoionization cross-section of the CO₂ molecule (squares, experimental results of ref 111); RCS-ADC(2) cross-sections obtained using respectively all the virtual orbitals from a GTO cc-pvDZ basis-set HF calculation (black line), the energy-truncated virtual orbitals from a GTO cc-pVTZ basis-set HF calculation (red line), and all the virtual orbitals from a GTO cc-pVTZ basis-set HF calculation (blue line). Bottom panel: B-spline RCS-ADC(2)_x total photoionization cross-section of the CO₂ molecule (squares, experimental results of ref 111); RCS-ADC(2)_x cross-sections obtained using respectively all the virtual orbitals from a GTO cc-pvDZ basis-set HF calculation (black line), the energy-truncated virtual orbitals from a GTO cc-pVTZ basis-set HF calculation (red line), and all the virtual orbitals from a GTO cc-pVTZ basis-set HF calculation (blue line).

Details of the calculations, i.e., the total dimension of the ADC(1), RCS-ADC(2), and RCS-ADC(2)_x matrices (eq 36), the size of the RCS configurations sub-block, and the number of ionic states included in the RCS-ADC cross-section calculations presented in this work, are reported in Table 1.

Regarding the computational cost of the calculations presented here, the scaling for the calculation of the RCS Hamiltonian $H_{\text{RCS}}^{\text{ADC}(n)}$ and dipole $D_{\text{RCS}}^{\text{ADC}(2)x}$ blocks (in eq 36 and eq 55, respectively), with respect to the number of RCS orbitals N_g used, remains the same as in conventional ADC and is respectively N_g^4 and N_g^5 . When increasing the size of the system, for example from N_2 to a larger polyatomic molecule, the number of polycentric GTO functions included in the RCS will increase linearly with respect to the number of atoms (a typical polycentric GTO basis set, such as cc-pVTZ, contains 30 functions per atom). The size of the RCS Hamiltonian and dipole blocks will also increase accordingly, especially due to the higher number of included 2h2p RCS configurations.

However, in typical RCS-ADC calculations aiming at describing molecular ionization, the size of the full B-spline orbital basis, which includes both the OCE and LCAO expansions and describes both the RCS and IS, is much larger with respect to the RCS one, the typical number of orbitals in the IS ($N_{\text{IS}} = N_b - N_g$) easily exceeding a few thousands ($N_{\text{IS}} \sim 2000$ per IRREP, in the CO₂ calculation reported in this work) and typically satisfying the relation $N_{\text{IS}} > N_g$ at least for molecules with up to ~ 20 atoms. Therefore, the most expensive part of the computation consists of the calculation of the interchannel coupling terms in eq 42 between the $\hat{c}_\mu^\dagger |\Psi_n^{N-1}\rangle$ product states. Although, when increasing the size of the molecular system, the size of the full B-spline orbital basis to be used will also increase (for instance due to the necessary increase in the number of off-center B-splines) in order to account for the larger RCS, the size of the IS, by which the description of the photoelectron wave function outside of the molecular region is exclusively done, will remain more or less the same. Therefore, for a fixed number of ionic channels included in the calculation, the computation of the matrix elements of eq 42 will scale approximately as N_{RCS}^2 , where N_{RCS} is the average number of 1h and 2h1p RCS-ADC configurations in the expansion of the ADC ionic states $|\Psi_n^{N-1}\rangle$ included. In the calculations reported in this work the CPU time increased from approximately 1 h when computing the RCS-ADC(2)_x interchannel coupling block (eq 42) for the HF molecule ($N_{\text{RCS}} = 306$) to approximately 5 h for the case of the CO₂ molecule ($N_{\text{RCS}} = 702$). The total CPU time, including the Hamiltonian diagonalization, required for the CO₂ cross-section calculations presented in this work is of the order of a few minutes for the ADC(1) method and of approximately 36 h for the RCS-ADC(2) level of theory, when using 80 computational cores.

The B-spline ADC(1) results reproduce accurately the overall behavior of the cross-section, although they shift the position of the cross-sections peaks to higher energy with respect to the experimental one. This is a general feature of the ADC(1) level of theory that overestimates the final state energies. Note that although the CIS and ADC(1) methods give the same energies for the excited states of the molecule, the B-spline CIS results, differently from ADC(1), considerably overestimate the value of the cross-section over the entire energy range. The difference between the CIS and ADC(1) results is given by the modified transition dipole moments between the ground state and the complete manifold of singly excited states, which, within ADC(1), include the contribution of double excitations in the ground state, evaluated at the MP2 level. The MP2-corrected ADC(1) transition dipole moments are indeed expressed as

Table 1. Details of the RCS-ADC Stieltjes Photoionization Cross-Section Calculations for Which the Converged Cross-Sections Presented in Figures 1, 2, 3, and 4 Have Been Obtained^a

molecule	ADC(1) tot dims	RCS-ADC(2) tot dims	RCS-ADC(2) _x tot dims	RCS-ADC(2) _x RCS-block dims
HF	($N_{\text{ion}} = 4$) 6513	($N_{\text{ion}} = 10$) 21486	($N_{\text{ion}} = 15$) 29564	5301
H ₂ O	($N_{\text{ion}} = 4$) 6868	($N_{\text{ion}} = 16$) 36390	($N_{\text{ion}} = 25$) 51729	9204
N ₂	($N_{\text{ion}} = 5$) 4339	($N_{\text{ion}} = 20$) 26310	($N_{\text{ion}} = 32$) 36404	8955
CO ₂	($N_{\text{ion}} = 8$) 14796	($N_{\text{ion}} = 22$) 62730	($N_{\text{ion}} = 33$) 81816	24598

^aThe highest value (among the different symmetry spaces) of the total dimensions of the ADC(1), cc-pVTZ RCS-ADC(2), and cc-pVTZ RCS-ADC(2)_x matrices. In the last column the dimensions of the RCS configurations sub-block are also shown. In parentheses the number of ADC open ionic channels included in each calculation (see eq 35) are also shown.

$$\langle \tilde{\Psi}_i^a | \hat{D} | \Psi_0 \rangle = d_{ai} + \sum_b \sum_j \frac{\langle ballji \rangle}{\epsilon_j + \epsilon_i - \epsilon_b - \epsilon_a} d_{jb} \quad (68)$$

Despite this being the only MP2 correction of ADC(1), it affects exactly those matrix elements that embody the ground state photoionization and considerably improves the CIS results.

Note that although the overall behavior of the cross-section is correctly reproduced by the ADC(1) as well as the RCS-ADC(2) and RCS-ADC(2) x theoretical curves, better agreement with the experiment is obtained within the second-order methods. In general, in the energy region close to the first ionization threshold, the B-spline RCS-ADC results have comparable accuracy to the ones obtained in ref 70, where the GTO-based conventional ADC approach was used. However, if one looks carefully at the high-energy tail of the cross-section, it is possible to see that much better agreement with the experiment is obtained within the new B-spline basis-set implementation. This is due to the fact that B-spline calculations show very good stability of the Stieltjes orders,⁸⁸ in contrast to the GTO calculation in which no convergence of the SI procedure can be detected.

The accuracy of the SI technique is usually limited by both the ability of the chosen L^2 basis set to represent strongly oscillating continuum functions and by the numerical instability of the computational algorithm of the Stieltjes–Chebyshev moment theory. The B-spline basis set removes the first limitation, accurately representing the continuum functions within the interaction volume as well as providing a much higher density of states, in the energy domain, in the continuum discretization, making it possible to correctly reproduce many high-order spectral moments in the SI framework.¹⁰⁶ The immediate consequence of this is that a considerable number of Stieltjes orders become reliable, and, as a result, the low resolution of the GTO converged Stieltjes orders has been considerably improved, over the entire energy range, by the B-spline results. Indeed, while in the GTO calculations of ref 70 the relevant stable Stieltjes orders used to obtain the interpolated cross-section were 6–10, the B-spline Stieltjes orders show much higher stability and the cross-sections reported in this work have been obtained by interpolating the Stieltjes points accumulating from the Stieltjes orders 23–32. However, this resolution is in general not yet enough to describe very sharp spectral features; i.e., the inability of the Stieltjes–Chebyshev moment theory to reproduce features resulting from a series of closely lying autoionization resonances (typically in the energy region close to the ionization threshold) persists also in the B-spline implementation.

Figure 1 shows the B-spline CIS, ADC(1), RCS-ADC(2), and RCS-ADC(2) x theoretical photoionization cross-sections of the HF molecule in the energy range from the first ionization threshold to 140 eV, together with the experimental result of Carnovale et al.¹⁰⁷ The origin of the single-center expansion has been placed on the F atom. The maximum value of angular momentum $l_{\max}^{\text{OCE}}$ used in the partial wave OCE expansion of the continuum orbitals is $l_{\max}^{\text{OCE}} = 8$, which provides convergent results for the calculated cross-sections in the entire photon energy range explored. The results show that the higher order schemes lead to a much improved peak position with respect to the ADC(1) calculation. Furthermore, differently from the GTO-based conventional ADC results,⁷⁰ they also reproduce precisely the cross-section peak height.

However, one may note that the cross-section shoulder above 40 eV is still not reproduced correctly.

Figure 2 shows the B-spline CIS, ADC(1), RCS-ADC(2), and RCS-ADC(2) x theoretical photoionization cross-sections of the H₂O molecule in the energy range from the first ionization threshold to 140 eV, together with the experimental results of Tan et al.¹⁰⁸ and the ones of Haddad and Samson.¹⁰⁹ The origin of the single-center expansion has been placed on the O atom. The maximum value of angular momentum $l_{\max}^{\text{OCE}}$ used in the partial wave OCE expansion of the continuum orbitals is $l_{\max}^{\text{OCE}} = 8$, which provides convergent results for the calculated cross-sections in the entire photon energy range explored. The agreement between the experimental and the theoretical cross-sections improves dramatically from ADC(1) to RCS-ADC(2), but not nearly as much when going from RCS-ADC(2) to the RCS-ADC(2) x level of theory. The RCS-ADC(2) and the RCS-ADC(2) x results essentially coincide with the experimental data of ref 109, and they also reproduce the sharp feature at 15 eV, which was missing in the GTO-ADC calculations.⁷⁰

Figure 3 shows the B-spline CIS, ADC(1), RCS-ADC(2), and RCS-ADC(2) x theoretical photoionization cross-sections of the N₂ molecule in the energy range from the first ionization threshold to 140 eV, together with the experimental result of Samson et al.¹¹⁰ The origin of the single-center expansion has been placed on the center of mass of the system (on the middle of the N–N bond). The maximum value of angular momentum $l_{\max}^{\text{OCE}}$ used in the partial wave OCE expansion of the continuum orbitals is $l_{\max}^{\text{OCE}} = 8$, which provides convergent results for the calculated cross-sections in the entire photon energy range explored. One can see that the agreement between the experimental and the theoretical cross-sections improves with the order of the ADC scheme. Both RCS-ADC(2) and RCS-ADC(2) x results are in impressive agreement with the experiment. Moreover, as a result of the better description of the ionic states with strong shakeup character given by the RCS-ADC(2) x method compared to the RCS-ADC(2) one, the small peak in the 18–27 eV energy range plateau, which appears at 23.5 eV in the experimental cross-section and is missed by both the ADC(1) and RCS-ADC(2) results, is also qualitatively reproduced by the RCS-ADC(2) x result, its position being shifted to higher energy by 2 eV with respect to the experimental one.

Figure 4 shows the B-spline CIS, ADC(1), RCS-ADC(2), and RCS-ADC(2) x theoretical photoionization cross-sections of the CO₂ molecule in the energy range from the first ionization threshold to 140 eV, together with the experimental result of Chan et al.¹¹¹ The origin of the single-center expansion has been placed on the C atom, which coincides with the center of mass of the system. The maximum value of angular momentum $l_{\max}^{\text{OCE}}$ used in the partial wave OCE expansion of the continuum orbitals is $l_{\max}^{\text{OCE}} = 13$, which provides convergent results for the calculated cross-sections in the entire photon energy range explored. One can see that the agreement between the experimental and the theoretical cross-sections improves with the order of the ADC scheme. Differently from the GTO-based conventional ADC calculations,⁷⁰ the double narrow peak at near 20 eV is now correctly reproduced by the Stieltjes-imaging B-spline RCS-ADC results. Moreover, the oscillations present in the experimental cross-section in the 30–60 eV range are fully reproduced by the second-order RCS-ADC calculations. More specifically, it is possible to observe that the RCS-ADC(2) and

RCS-ADC(2) x results correctly reproduce the energy position as well as the shape of both the first plateau around 35 eV, which was partially reproduced by both GTO-based second-order ADC(2) and ADC(2) x methods (see ref 70), and the second plateau at around 55 eV, which was missing altogether in both the first-order ADC(1) and second-order ADC(2) GTO results.

In Figure 5 the RCS-ADC(2) and RCS-ADC(2) x cross-sections of the CO₂ molecule obtained using different RCSs are showed. The RCSs used consist respectively of all the virtual orbitals from a GTO cc-pVDZ basis-set HF calculation, the energy-truncated ($\epsilon^{\text{cut}} = 6$ au) virtual orbitals from a GTO cc-pVTZ basis-set HF calculation, and all of the virtual orbitals from a GTO cc-pVTZ basis-set HF calculation. As one can observe, the use of a TZ basis leads to a significant improvement in the accuracy of the cross-section with respect to the use of DZ ones. Indeed, the cc-pVDZ RCS-ADC calculations tend to underestimate the value of the cross-section close to the ionization thresholds and overestimate it at higher energies. Conversely, very good agreement with the experimental results was achieved by using the cc-pVTZ basis sets.

The accuracy of bound-continuum transitions calculated using cc-pVTZ GTO-expanded RCSs can also be observed by comparing the conventional B-spline ADC(1) results for the total photoionization cross-sections with the B-spline RCS-ADC(1) ones. Within the RCS-ADC(1) approach, the excited state energies are equivalent to the conventional ADC(1) ones, while the modified transition dipole moments from the RCS ground state to the complete manifold of excited and ionized states now reads as

$$\langle \tilde{\Psi}_i^a | \hat{D} | \Psi_0 \rangle = d_{ai} + \sum_{\alpha \in \text{RCS}} \sum_j \frac{\langle \alpha | \text{all} | j \rangle}{\epsilon_j + \epsilon_i - \epsilon_\alpha - \epsilon_a} d_{ja} \quad (69)$$

where the virtual summation only runs over RCS virtual orbitals (χ_α), instead of the entire virtual space. In Figure 6,

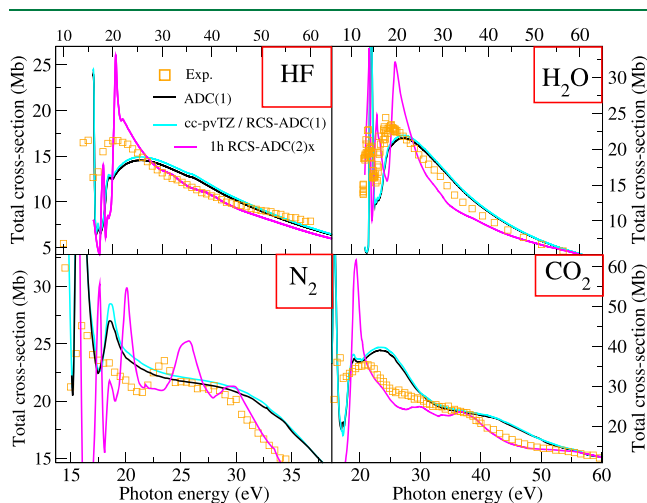


Figure 6. Total photoionization cross-section of the HF (a), H₂O (b), N₂ (c), and CO₂ (d) molecule (black line, B-spline ADC(1) Stieltjes result; blue line, B-spline RCS-ADC(1) Stieltjes result; magenta line, B-spline RCS-ADC(2) x Stieltjes cross-section obtained using simplified 1h representation for the ionic states in the CC expansion of eq 35). All the RCS-ADC(1) and RCS-ADC(2) x cross-sections are obtained using a RCS consisting of all the virtual orbitals from a GTO cc-pVTZ HF calculation.

both the ADC(1) and RCS-ADC(1) results for the total photoionization cross sections of HF, H₂O, N₂ and CO₂ are showed. As can be seen, the two methods give practically the same result, giving strong indication that restricting the correlation space to cc-pVTZ calculated virtual orbitals is a very good approximation for calculating bound-continuum transitions.

Furthermore, within the RCS-ADC approaches, we should note the importance of treating the RCS bound N -electron and $(N - 1)$ -electron ADC states at the same level of ab initio theory. The consistency of their description is essential to obtain quantitatively as well as qualitatively accurate results. In order to demonstrate this, in Figure 6 I also show the total photoionization cross-sections of HF, H₂O, N₂, and CO₂, calculated using a simplified RCS-ADC(2) x approach where the open ionic channels in the CC expansion of eq 35 are described as simple 1h configurations instead of ADC(2) x ionic states. As one can see, the simplified RCS-ADC(2) x results fail to correctly reproduce the value of the cross-sections in the outer-valence energy region, artificially predicting spurious sharp features in the vicinity of the main peak, which are not present both in the full RCS-ADC(2) x results and in the experimentally measured cross-sections.

Finally, it is interesting to analyze the relative deviations of the calculated cross-sections from the experimental ones, as a function of the photon energy and of the different B-spline RCS-ADC(n) ab initio method used. Figure 7 shows that both

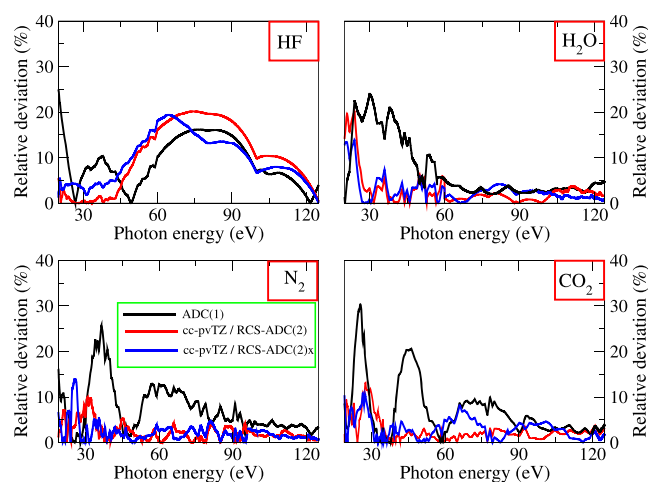


Figure 7. Relative deviations of the RCS-ADC Stieltjes photoionization cross-sections from the experimental results for each of the four molecules calculated as a function of the energy in the energy range of ionization threshold to 130 eV (black line, RCS-ADC(1) Stieltjes result; red line, RCS-ADC(2) Stieltjes cross-section; green line, RCS-ADC(2) x Stieltjes cross-section).

RCS-ADC(2) and RCS-ADC(2) x methods lead to impressive agreement with experiment, with the relative deviations being, for the H₂O, N₂, and CO₂, around 5% over the entire photon-energy region from the first ionization potentials up to 130 eV. As it possible to see, for the case of the HF molecule, increasing the level of correlation description from ADC(1) to RCS-ADC(2) x does not significantly improve the description of the cross-section shoulder above 40 eV. However, the theoretical results presented here are in good agreement with the previous GTO based ADC calculations,⁷⁰ with the theoretical multichannel random phase approximation (MC-

RPA) results of Cacelli et al.¹¹² as well as the coupled cluster results of Cukras et al.⁷¹ This may call for a revision of the experimental cross-section for HF. Therefore, the new RCS-ADC approaches maintain the same level of accuracy in describing the electron-correlation effects in molecular photoionization as the conventional ADC ones.⁷⁰ Moreover, the dramatically improved feasibility of their multicenter B-spline implementation, with respect to the conventional ADC approaches, allows one to keep this level of accuracy in the description of bound-continuum transitions over a very much extended energy range, not limited anymore to the energy region close to the first ionization threshold. Indeed, the precision deterioration at higher energies typical of the GTO calculations⁷⁰ is not present any more, the average error stabilizing around 5% for both methods.

CONCLUSIONS

In this work I presented a new version of the ab initio ADC schemes for the polarization propagator, explicitly designed to accurately and efficiently describe molecular ionization. I adapted the ADC(*n*) schemes to problems where the accurate description of both the ionization continuum and the complicated short-range structure of correlated electronic wave functions is required, by exploiting the restricted correlation space ansatz. The RCS version of the ADC methods up to second order of perturbation theory has been derived via the intermediate state representation (ISR) and implemented in the multicenter B-spline basis set for the electronic continuum.

Remarkably, a general close-coupling structure of the RCS-ADC many-electron wave function, connecting the *N*-particle to the (*N* − 1)-particle ADC intermediate states, emerges naturally as a nontrivial result of the RCS ansatz and without the need for an ad hoc inclusion of the physically relevant ionic states. The structure of eq 34, which links the RCS-ADC(*n*) wave function of the *N*-electron system to the respective ADC(*n*) (*N* − 1)-electron eigenstates, is general and applies to every order *n* of the ADC hierarchy. As a result, within the RCS-ADC(*n*) schemes for molecular ionization, electron correlation is included in the ab initio description of multichannel molecular ionization at the systematically improvable level of theory. Moreover, overcoming the practical limitations of the conventional ADC approaches, the newly introduced second-order RCS-ADC schemes prove to be particularly manageable from a computational point of view, which makes their simultaneous implementation in the multicenter B-spline basis set and application to polyatomic molecules entirely feasible.

The quality and robustness of the new RCS-ADC(*n*) approaches and the accuracy of the implementation were verified by performing a series of total photoionization cross-section calculations on a test set of molecules. The results presented in this work demonstrate that the new RCS variants of the ADC methods maintain the same level of accuracy in describing the electron correlation intrinsic to molecular ionization as the conventional ADC ones. The dramatically improved feasibility of their multicenter B-spline implementation, with respect to the conventional ADC approaches, allows for an accurate description of bound-continuum transitions over a very much extended energy range, without being limited anymore, as in previous GTO implementations, to a good description of the outer-valence energy region close to the molecular ionization threshold. The excellent agreement of the

results presented here with existing accurate benchmarks, both experimental and theoretical GTO-ADC, shows that the RCS-ADC schemes overcome the difficulties of extending the conventional B-spline ADC methods to correlated ionization dynamics in molecular systems, while encompassing the level of correlation description of standard ADC schemes and the optimal representation of the electronic continuum. Thus, I demonstrated that the RCS versions of the ADC schemes are optimal and quantitatively accurate methods for studying multichannel ionization in closed-shell molecular systems.

Future applications of the RCS-ADC approaches will focus on dynamic calculations. The RCS-ADC(*n*) schemes lend themselves naturally to the disentanglement of the *N*-electron wave function structure, across a continuous range of energies, into the asymptotic wave function of the ejected electron and the states of the parent molecular ion. This feature is essential in order to analyze the post-ionization wave function and makes the RCS-ADC methods extremely promising approaches for studying the ionization dynamics by solving the TDSE for molecules interacting with laser pulses. Applications of the B-spline RCS-ADC method to bigger polyatomic molecules as well as to the study of the time-dependent many-electron dynamics occurring during molecular ionization is currently in progress and will be reported in following works.

APPENDIX

Here, the derivations of eq 16 and eq 28 are outlined. The ISR-ADC two-step orthonormalization procedure is now applied to the case of the second (2h2p) RCS-ADC excitation class. The doubly excited precursor states with respectively one and two electrons promoted to IS orbitals can be written as

$$\begin{aligned} |\Psi_{\mu,\beta kl}^\# \rangle &= \hat{c}_\mu^\dagger \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle - \sum_{\nu i} |\tilde{\Psi}_{\nu,i}^N \rangle \langle \tilde{\Psi}_{\nu,i}^N | \hat{c}_\mu^\dagger \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle \\ &= \hat{c}_\mu^\dagger \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle - \sum_{\nu i} \hat{c}_\nu^\dagger |\tilde{\Psi}_i^{N-1} \rangle \langle \tilde{\Psi}_i^{N-1} | \hat{c}_\mu^\dagger \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle \end{aligned} \quad (70)$$

$$|\Psi_{\mu\nu,kl}^\# \rangle = \hat{c}_\mu^\dagger \hat{c}_\nu^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle = \hat{c}_\mu^\dagger \hat{c}_\nu^\dagger |\Psi_{k'l'}^{N-2\#} \rangle \quad (\mu > \nu) \quad (71)$$

where $|\Psi_{k'l'}^{N-2\#} \rangle = \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle$ are correlated (*N* − 2)-electron states of ISR-ADC for doubly ionized states.

In order to obtain eqs 70 and 71, the orthogonality relations between the IS 2h2p states and the RCS ground state

$$\begin{aligned} \langle \Psi_0^{\text{RCS}} | \hat{c}_\mu^\dagger \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle &= 0 \\ \langle \Psi_0^{\text{RCS}} | \hat{c}_\mu^\dagger \hat{c}_\nu^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle &= 0 \end{aligned} \quad (72)$$

which follow from the RCS ansatz of eq 9, have been used. Using the anticommutation relations of eq 8 and the RCS ansatz (eq 9), the precursor states of eq 70 can be written as

$$\begin{aligned} |\Psi_{\mu,\beta kl}^\# \rangle &= \hat{c}_\mu^\dagger \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle - \sum_{\nu i} \delta_{\mu\nu} \hat{c}_\nu^\dagger |\tilde{\Psi}_i^{N-1} \rangle \langle \tilde{\Psi}_i^{N-1} | \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle \\ &= \hat{c}_\mu^\dagger \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle - \sum_i |\tilde{\Psi}_i^{N-1} \rangle \langle \tilde{\Psi}_i^{N-1} | \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle \\ &= \hat{c}_\mu^\dagger |\Psi_{\beta kl}^{N-1\#} \rangle \end{aligned} \quad (73)$$

where $\hat{c}_\mu^\dagger |\Psi_{\beta kl}^{N-1\#} \rangle = \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle - \sum_i |\tilde{\Psi}_i^{N-1} \rangle \langle \tilde{\Psi}_i^{N-1} | \hat{c}_\beta^\dagger \hat{c}_k \hat{c}_l |\Psi_0^{\text{RCS}} \rangle$ are the correlated 2h1p precursor states of ISR-ADC for singly ionized states.^{98,99}

The doubly excited intermediate states are computed symmetrically orthonormalizing the corresponding precursor states of eqs 71 and 73. Using again the RCS ansatz (eq 9) and the anticommutation relations of eq 8, it is possible to see that the following matrix elements of the overlap matrix S^N , between N -electron second-excitation-class precursor states with different numbers of electrons excited in the IS space, are zero:

$$S_{\mu'v'k'l',\mu\beta kl}^N = 0, \quad S_{\mu'v'k'l',\alpha\beta kl}^N = 0, \quad S_{\mu'v'k'l',\alpha\beta kl}^N = 0 \quad (74)$$

The only contributions to the overlap matrix S^N come from matrix elements between doubly excited states with the same number of electrons excited in the IS space, namely,

$$S_{\mu'v'k'l',\mu\beta kl}^N = \langle \Psi_{\beta'k'l'}^{N-1\#} | \hat{c}_\mu^\dagger \hat{c}_\mu^\dagger | \Psi_{\beta kl}^{N-1\#} \rangle = \delta_{\mu\mu'} S_{\beta'k'l',\beta kl}^{(N-1)} \quad (75)$$

$$S_{\mu'v'k'l',\mu\nu kl}^N = \langle \Psi_0^{\text{RCS}} | \hat{c}_l^\dagger \hat{c}_k^\dagger \hat{c}_{\nu'}^\dagger \hat{c}_{\mu'}^\dagger \hat{c}_\mu^\dagger \hat{c}_k^\dagger \hat{c}_l | \Psi_0^{\text{RCS}} \rangle \quad (76)$$

Using the anticommutation relations of eq 8 and the RCS ansatz (eq 9), the expression in eq 76 becomes

$$\begin{aligned} S_{\mu'v'k'l',\mu\nu kl}^N &= \delta_{\mu\mu'} \delta_{\nu\nu'} \langle \Psi_0^{\text{RCS}} | \hat{c}_l^\dagger \hat{c}_k^\dagger \hat{c}_k \hat{c}_l | \Psi_0^{\text{RCS}} \rangle \\ &\quad - \delta_{\mu\nu'} \delta_{\nu\mu'} \langle \Psi_0^{\text{RCS}} | \hat{c}_l^\dagger \hat{c}_k^\dagger \hat{c}_k \hat{c}_l | \Psi_0^{\text{RCS}} \rangle \\ &= (\delta_{\mu\mu'} \delta_{\nu\nu'} - \delta_{\mu\nu'} \delta_{\nu\mu'}) S_{k'l',kl}^{(N-2)} \end{aligned} \quad (77)$$

In eqs 75 and 77), $S^{(N-1)}$ and $S^{(N-2)}$ are respectively the overlap matrices in the spaces of $(N-1)$ -electron 2h1p and $(N-2)$ -electron 2h ISR-ADC precursor states.

Finally, the IS doubly excited RCS-ADC states are obtained as

$$\begin{aligned} |\tilde{\Psi}_{\mu,\beta kl}\rangle &= \sum_{\mu'v'k'l'} \hat{c}_\mu^\dagger |\Psi_{\beta'k'l'}^{N-1\#}\rangle \delta_{\mu\mu'} S_{\beta'k'l',\beta kl}^{(N-1)^{-1/2}} \\ &= \hat{c}_\mu^\dagger \sum_{\beta'k'l'} |\Psi_{\beta'k'l'}^{N-1\#}\rangle S_{\beta'k'l',\beta kl}^{(N-1)^{-1/2}} \\ &= \hat{c}_\mu^\dagger |\tilde{\Psi}_{\beta kl}^{N-1}\rangle \end{aligned} \quad (78)$$

$$\begin{aligned} |\tilde{\Psi}_{\mu\nu,kl}\rangle &= \frac{1}{2} \sum_{\mu'v'k'l'} |\Psi_{\mu'v',k'l'}^{\#}\rangle (S^{N-1/2})_{\mu'v'k'l',\mu\nu kl} \\ &= \frac{1}{2} \sum_{\mu'v'k'l'} \hat{c}_\mu^\dagger \hat{c}_{\nu'}^\dagger |\Psi_{k'l'}^{N-2\#}\rangle (\delta_{\mu\mu'} \delta_{\nu\nu'} - \delta_{\mu\nu'} \delta_{\nu\mu'}) S_{k'l',kl}^{(N-2)^{-1/2}} \\ &= \hat{c}_\mu^\dagger \hat{c}_{\nu'}^\dagger |\tilde{\Psi}_{kl}^{N-2}\rangle \end{aligned} \quad (79)$$

where the multiplicative factor $\frac{1}{2}$ in eq 79 takes into account that only summation over $\mu' > \nu'$ is actually performed.

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Notes

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